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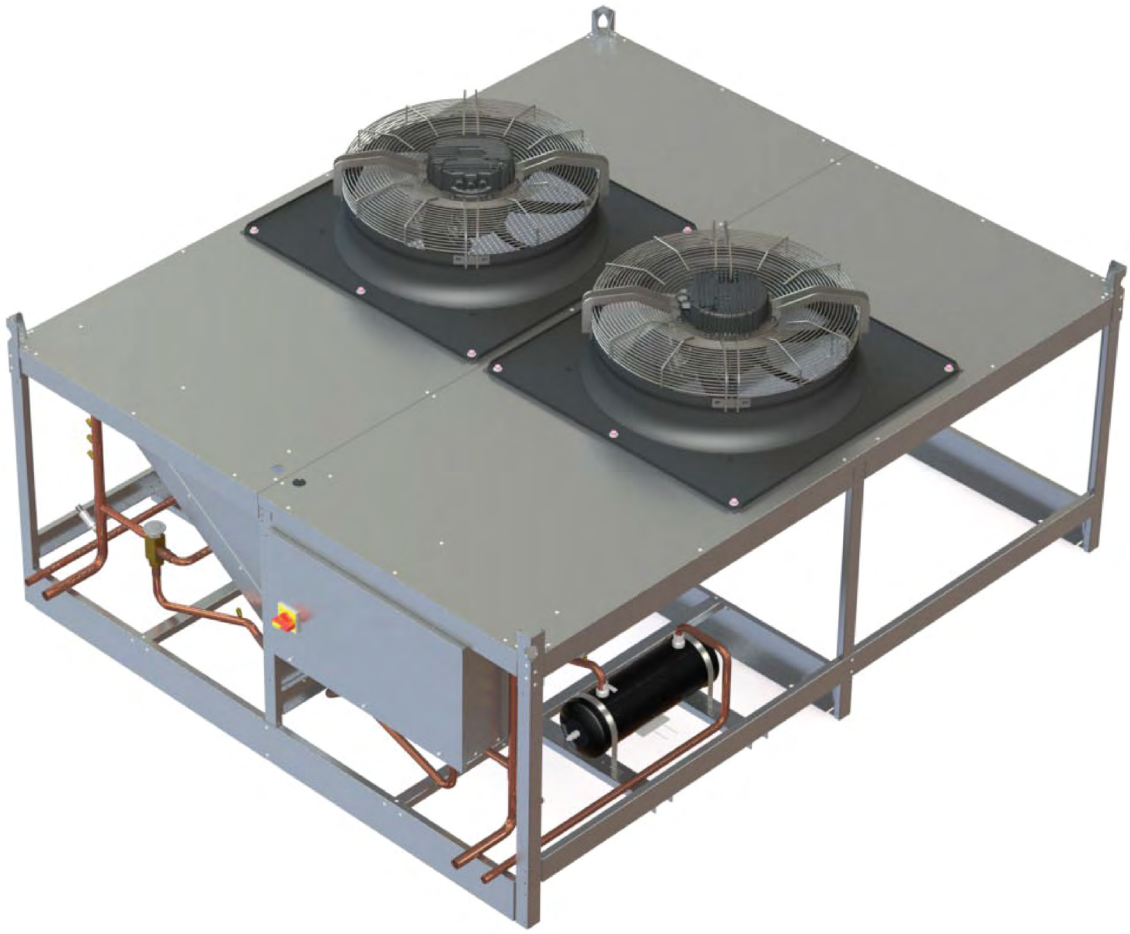
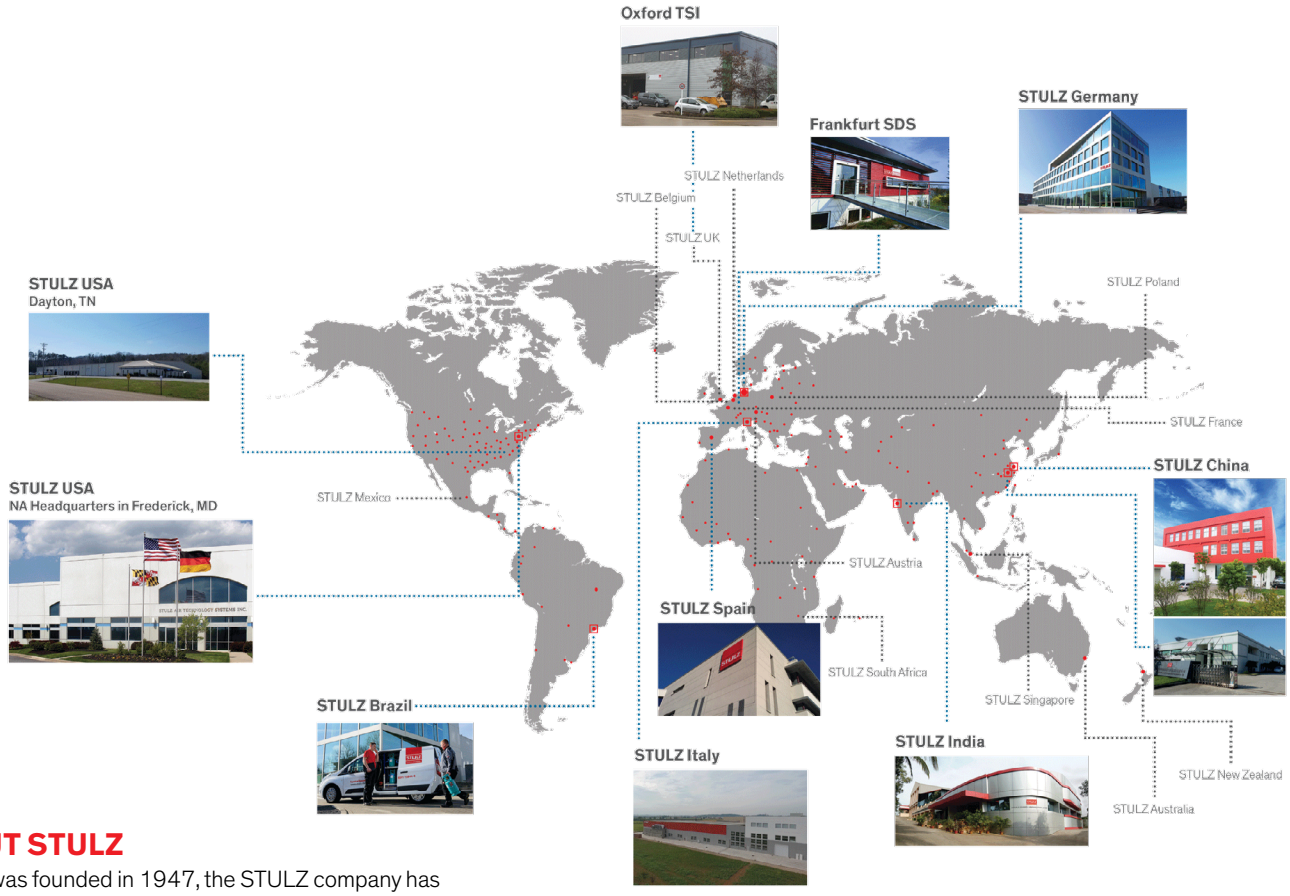


Heat Rejection

Engineering Manual

Air Cooled Condensers
For use with STULZ Precision Air Conditioners (60 Hz Data)



ABOUT STULZ

Since it was founded in 1947, the STULZ company has evolved into one of the world's leading system suppliers of computer room air conditioners and air handlers. STULZ has seen continual international expansion of its precision cooling business, specializing in temperature and humidity control for data centers and telecommunications installations.

STULZ has ten production plants (two in Germany, two in the USA, Italy, Great Britain, Spain, two in China, Brazil and India), and twenty subsidiaries (in Germany, USA, France, Italy, Great Britain, Belgium, Brazil, the Netherlands, New Zealand, Mexico, Austria, Poland, Spain, Singapore, China, India, Indonesia, South Africa, Sweden, and Australia).

STULZ works with sales and service partners in over 140 other countries, and therefore boasts an international network of cooling specialists.

The benchmark of flexibility and energy efficiency

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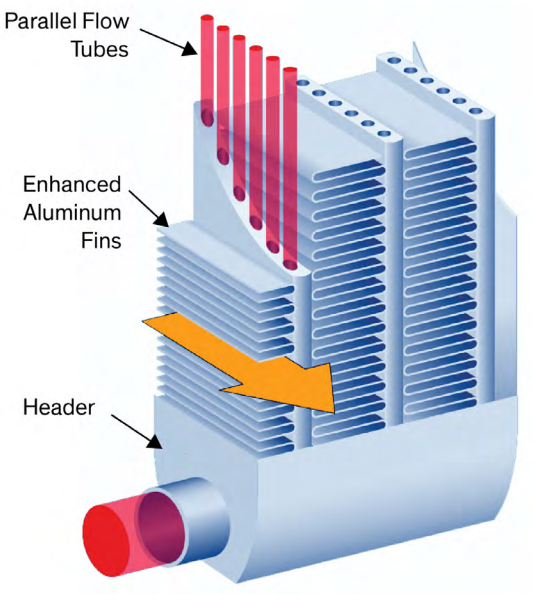
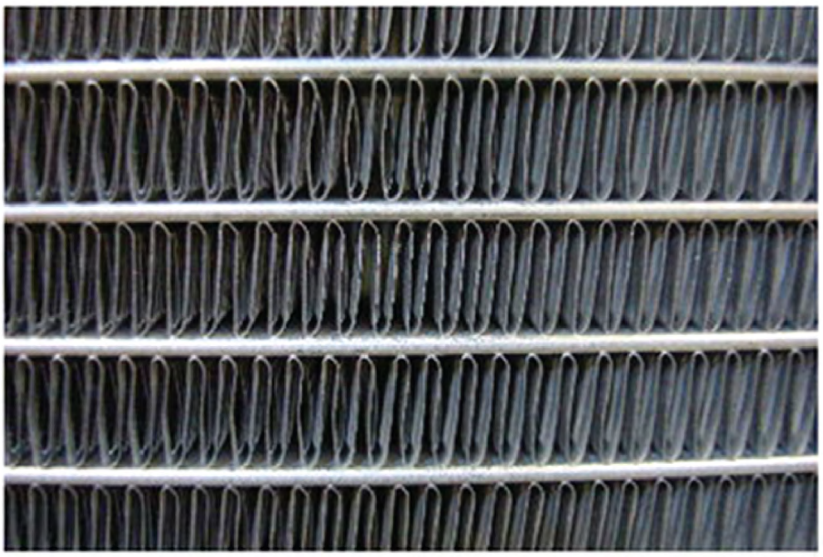
SCS-MC Offers Outstanding Advantages



Micro-Channel Design



Micro-Channel Coil Profile View

Micro-Channel Condenser Features

Micro-Channel Condensers are high efficiency state-of-the-art condensers with high total heat rejection, reduced weight, a smaller footprint, and improved Standard Coefficient of Performance (SCOP).

Thermal Performance

- Improved heat rejection compared to a standard three-row, Al/Cu coil
- Low ambient temperature control options (-30°F)
- Factory installed and piped receiver

Efficiency

- Uses highly efficient axial fans with AC and EC motor configurations
- Integrated control box with main disconnect and motor protection
- Various fan speed control options based on refrigerant pressure

Build Quality

- Structurally robust: more durable, even with significant weight reduction
- Superior corrosion resistance due to all aluminum construction (cabinet and fins)
- Aluminum brazed fin construction superior to Al/Cu (two metal) coils

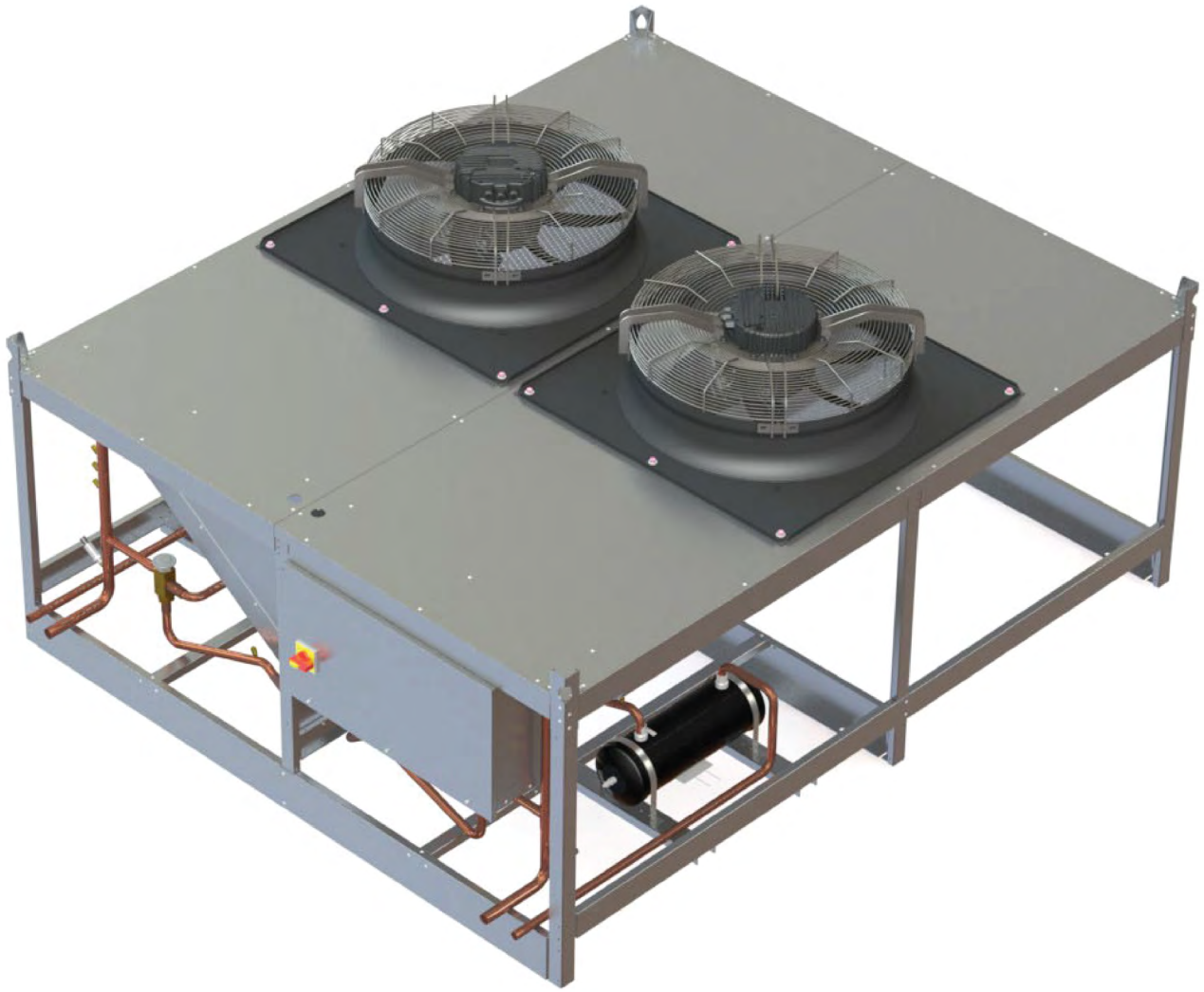
Environmentally Friendly

- R410A and R407C refrigerant
- Smaller volume, up to 20% less refrigerant use at the same capacity

Designed for Serviceability

- Easily cleaned and field repaired
- Optional pre-filters for easy maintenance

SCS-MC Micro-Channel Outdoor Air Cooled Condensers



Model SCS-MC-142-D Shown

See model nomenclature on next page.

1. Nomenclature

SCS-MC-XXX-XXX-XX					
SCS-MC	**Capacity (kW)**	**Number of Coolant Circuits**	**Head Pressure Controls & Fan Options**	**Receiver Options**	**Refrigerant**
SCS-MC = Condenser with Micro-Channel Coils	015 018	S = Single	AA = Fan Cycle Control, AC Fans SA = Variable Speed Control, EC and AC Fans EC = Intelligent Control, Electronically Commutated Fans LN = Low Noise Intelligent Control, Electronically Commutated Fans	O = Standard F = Flooded Head Pressure Control with Receiver	1 = R407C 2 = R410A
	031 035 056 071 111	S = Single D = Dual			
	128 142 223 264 334	D = Dual			

Features

- Micro-Channel Coil
- Provides better heat rejection
- Less refrigerant used in the system
- Smaller footprint
- Less weight
- R410A or R407C refrigerant
- All aluminum cabinet construction
- Uses highly efficient axial fans with AC and EC motor configurations
- Integrated control box with main disconnect and motor protection
- Various fan speed control options based on refrigerant pressure
- Single and dual circuit configurations
- Low ambient temperature control options (-30°F)
- Factory installed and piped receiver

2. Performance Data

2.1. AA, SA, & EC Models

Model	Flow Rate	Total Heat Rejection w/ R410A (MBH)			Total Heat Rejection w/ R407C (MBH)			Noise	Refrigerant Connections (in)	
	(CFM)	@ 95°F	@ 105°F	@ 115°F	@ 95°F	@ 105°F	@ 115°F	(dBA)	Hot Gas Line	Liquid Line
SCS-MC-015-S	2,525	54	40	26	52	39	26	73	1/2	1/2
SCS-MC-018-S	2,775	67	50	33	65	49	33	73	1/2	1/2
SCS-MC-031-S	4,750	113	80	53	103	77	52	79	5/8	5/8
SCS-MC-031-D	4,750	113	80	53	103	77	52	79	5/8	5/8
SCS-MC-035-S	5,100	125	94	62	121	90	61	79	7/8	7/8
SCS-MC-035-D	5,100	125	94	62	121	90	61	79	5/8	5/8
SCS-MC-056-S	9,500	202	150	97	192	140	94	82	7/8	7/8
SCS-MC-056-D	9,500	202	150	97	192	140	94	82	5/8	5/8
SCS-MC-071-S	10,400	254	190	126	245	184	124	82	1 ⅛	7/8
SCS-MC-071-D	10,400	254	190	126	245	184	124	82	7/8	7/8
SCS-MC-111-S (1×2)	19,000	403	299	193	384	280	188	85	1 ⅛	7/8
SCS-MC-111-D (2×1)	19,000	403	299	193	384	280	188	85	7/8	7/8
SCS-MC-128-D	20,000	457	333	221	429	321	216	85	7/8	7/8
SCS-MC-142-D	20,800	509	380	253	490	367	249	85	1 ⅛	7/8
SCS-MC-223-D (2×2)	38,000	807	598	387	768	560	376	88	1 ⅛	7/8
SCS-MC-223-D (1×4)	38,000	807	598	387	768	560	376	88	1 ⅛	7/8
SCS-MC-264-D	40,000	932	684	454	881	659	445	88	1 ⅛	1 ⅛
SCS-MC-334-D	57,000	1,211	897	580	1,152	840	564	90	1 ⅛	1 ⅛

2.2. LN Models

Model	Flow Rate	Total Heat Rejection w/ R410A (MBH)			Total Heat Rejection w/ R407C (MBH)			Noise	Refrigerant Connections (in)	
	(CFM)	@ 95°F	@ 105°F	@ 115°F	@ 95°F	@ 105°F	@ 115°F	(dBA)	Hot Gas Line	Liquid Line
SCS-MC-015-SLN	1,135	30	22	N/A	28	21	N/A	58	1/2	1/2
SCS-MC-018-SLN	1,250	36	27	18	34	25	N/A	58	1/2	1/2
SCS-MC-031-SLN	2,140	58	44	29	55	42	28	63	5/8	5/8
SCS-MC-031-DLN	2,140	58	44	29	55	42	28	63	5/8	5/8
SCS-MC-035-SLN	2,295	66	49	33	63	47	31	63	7/8	7/8
SCS-MC-035-DLN	2,295	66	49	33	63	47	31	63	5/8	5/8
SCS-MC-056-SLN	4,275	109	82	55	104	78	52	66	7/8	7/8
SCS-MC-056-DLN	4,275	109	82	55	104	78	52	66	5/8	5/8
SCS-MC-071-SLN	4,680	134	101	67	127	96	64	66	1 ⅛	7/8
SCS-MC-071-DLN	4,680	134	101	67	127	96	64	66	7/8	7/8
SCS-MC-111-SLN (1×2)	8,550	219	164	109	208	155	104	69	1 ⅛	7/8
SCS-MC-111-DLN (2×1)	8,550	219	164	109	208	155	104	69	7/8	7/8
SCS-MC-128-DLN	9,000	243	182	121	231	173	115	69	7/8	7/8
SCS-MC-142-DLN	9,360	268	201	134	255	191	128	69	1 ⅛	7/8
SCS-MC-223-DLN (2×2)	17,100	437	327	218	416	311	207	72	1 ⅛	7/8
SCS-MC-223-DLN (1×4)	17,100	437	327	218	416	311	207	72	1 ⅛	7/8
SCS-MC-264-DLN	18,000	494	371	247	469	352	235	72	1 ⅛	1 ⅛
SCS-MC-334-DLN	25,650	656	491	327	623	466	311	74	1 ⅛	1 ⅛

2.3. Elevation Correction

Elevation above sea level has an effect on the performance of Air Cooled condensers. Multiply the rated condenser total heat rejection (THR) at sea level by the elevation correction factor in the table below when selecting the required condenser.

Elevation Correction Factors											
Elevation (ft)	1,000	2,000	3,000	4,000	5,000	6,000	8,000	10,000	12,000	14,000	16,000
Correction Factor	0.94	0.93	0.90	0.88	0.86	0.83	0.79	0.75	0.71	0.66	0.62

3. Technical Data

3.1. Dimensions and Weights

Model	Unit		Shipping	
	Overall Dimensions	Weight (lb)	Approximate Dimensions	Weight (lb)
SCS-MC-015-S	27.5" W x 29.7" H x 30.5" D	83	36" W x 42" H x 43" D	130
SCS-MC-018-S	37.0" W x 35.7" H x 30.5" D	104	45" W x 48" H x 43" D	175
SCS-MC-031-S	37.0" W x 35.7" H x 37.0" D	139	45" W x 48" H x 49" D	225
SCS-MC-031-D	37.0" W x 35.7" H x 44.6" D	139	45" W x 48" H x 57" D	225
SCS-MC-035-S	37.0" W x 35.7" H x 49.3" D	161	45" W x 48" H x 62" D	260
SCS-MC-035-D	37.0" W x 35.7" H x 56.9" D	161	45" W x 48" H x 69" D	260
SCS-MC-056-S	37.0" W x 35.7" H x 49.3" D	223	45" W x 48" H x 62" D	360
SCS-MC-056-D	37.0" W x 35.7" H x 56.9" D	223	45" W x 48" H x 69" D	360
SCS-MC-071-S	37.0" W x 35.7" H x 89.2" D	303	45" W x 48" H x 102" D	490
SCS-MC-071-D	37.0" W x 35.7" H x 96.9" D	303	45" W x 48" H x 109" D	490
SCS-MC-111-S (1×2)	37.0" W x 35.7" H x 89.2" D	402	45" W x 48" H x 102" D	650
SCS-MC-111-D (2×1)	73.8" W x 35.7" H x 49.3" D	415	82" W x 48" H x 62" D	670
SCS-MC-128-D	73.8" W x 35.7" H x 64.7" D	484	82" W x 48" H x 77" D	785
SCS-MC-142-D	73.8" W x 35.7" H x 89.2" D	560	82" W x 48" H x 102" D	1065
SCS-MC-223-D (2×2)	73.8" W x 35.7" H x 89.2" D	772	82" W x 48" H x 102" D	1240
SCS-MC-223-D (1×4)	37.0" W x 35.7" H x 176.8" D	772	45" W x 48" H x 189" D	1240
SCS-MC-264-D	73.8" W x 35.7" H x 129.2" D	898	82" W x 48" H x 142" D	1425
SCS-MC-334-D	73.8" W x 35.7" H x 129.2" D	1110	82" W x 48" H x 142" D	1625

3.2. Compatible A/C Models

SCS-MC Compatible A/C Models				
AA, SA & EC Models	R410A		R407C	
	@ 95°F	@ 105°F	@ 95°F	@ 105°F
SCS-MC-015	OHS-012 OHS-018 OHS-024	OHS-012 OHS-018 OHS-024	CCU/D-41 CCU/D-61 CCU/D-81 COS-024 OHS-012 OHS-018 OHS-024	CCU/D-41 CCU/D-61 COS-024 OHS-012 OHS-018 OHS-024
SCS-MC-018	OHS-032 OHS-040 CRS-042	COS-024 OHS-032 OHS-040	CCU/D-121 OHS-032 OHS-040	CCU/D-81 OHS-032 OHS-040
SCS-MC-031	OHS-042 OHS-048 OHS-060 OHS-084	CRS-042 OHS-048	CCU/D-171 CCU/D-201 COS-060 OHS-048 OHS-060	CCU/D-121 CCU/D-171 COS-060 OHS-048
SCS-MC-035	CFU/D-021 CRS-084 OHS-072	OHS-060	OHS-072	OHS-060
SCS-MC-056	COS-096 CFU/D-028 CFU/D-035 OHS-084 OHS-120	CFU/D-021 CFU/D-028 COS-096 CRS-084 OHS-084	COS-096 COS-120 OHS-084	CCU/D-201 COS-096 OHS-084
SCS-MC-071	COS-120	CFU/D-035 COS-120		COS-120
SCS-MC-111	CFU/D-053 CFU/D-070 CFU/D-088	CFU/D-053		
SCS-MC-128		CFU/D-070		
SCS-MC-142	CFU/D-105			
SCS-MC-223		CFU/D-088 CFU/D-105		

SCS-MC Compatible A/C Models				
LN Models	R410A		R407C	
	@ 95°F	@ 105°F	@ 95°F	@ 105°F
SCS-MC-015	OHS-012 OHS-018	OHS-012	CCU/D-41 OHS-012 OHS-018	OHS-012
SCS-MC-018	OHS-024	OHS-018	CCU/D-61 COS-024 OHS-024	CCU/D-41 OHS-018
SCS-MC-031	OHS-032 OHS-040	OHS-024 OHS-032 OHS-040	CCU/D-81 OHS-032 OHS-040	CCU/D-61 COS-024 OHS-024 OHS-032
SCS-MC-035	CRS-042		CCU/D-121	CCU/D-81 OHS-040
SCS-MC-056	OHS-048 OHS-060	CRS-042 OHS-048 OHS-060	CCU/D-171 CCU/D-201 COS-060 OHS-048 OHS-060	CCU/D-121 CCU/D-171 COS-060 OHS-048
SCS-MC-071	CFU/D-021 CRS-084 OHS-084		COS-096 OHS-084	CCU/D-201 OHS-060
SCS-MC-111	CFU/D-028 CFU/D-035	CFU/D-021 CFU/D-028 CFU/D-035 CRS-084 OHS-084	COS-120	COS-096 OHS-084
SCS-MC-128				COS-120
SCS-MC-142	CFU/D-053			
SCS-MC-223	CFU/D-070 CFU/D-088	CFU/D-053 CFU/D-070		
SCS-MC-264	CFU/D-105			
SCS-MC-334		CFU/D-088 CFU/D-105		

Compatible Refrigerant for DX Models	
R407C	R410A
OHS	CFS
COS	MiniSpace
	CRS

Key:
OHS = CeilAir Overheat Systems
CRS = CyberRow
CFU/D = CyberAir
CCU/D = MiniSpace
COS = CyberOne

3.3. Fan and Motor Data

SCS-MC Air Cooled Condensers - Single Circuit							
SCS-MC	**015**	**018**	**031**	**035**	**056**	**071**	**111**
-SAA Fan Cycling Control Package							
208–230/1/60, 277/1/60							
Fan RPM	1260	1260	1490	1490	N/A	N/A	N/A
Motor HP (Qty)	0.35 (1)	0.35 (1)	1.23 (1)	1.23 (1)	N/A	N/A	N/A
208–230/3/60							
Fan RPM	N/A	N/A	1445	1445	1585	1585	1585
Motor HP (Qty)	N/A	N/A	1.58 (1)	1.58 (1)	3.58 (1)	3.58 (1)	3.58 (2)
460/3/60, 575/3/60							
Fan RPM	1520	1520	1545	1545	1650	1650	1650
Motor HP (Qty)	0.31 (1)	0.31 (1)	1.76 (1)	1.76 (1)	3.90 (1)	3.90 (1)	3.90 (2)
-SSA Variable Fan Speed Control Package							
208–230/1/60, 277/1/60							
Fan RPM	1260	1260	1490	1490	N/A	N/A	N/A
Motor HP (Qty)	0.35 (1)	0.35 (1)	1.23 (1)	1.23 (1)	N/A	N/A	N/A
208–230/3/60							
Fan RPM	N/A	N/A	N/A	N/A	N/A	N/A	1500 (1) 1585 (1)
Motor HP (Qty)	N/A	N/A	N/A	N/A	N/A	N/A	3.85 (1) 3.58 (1)
460/3/60, 575/3/60							
Fan RPM	N/A	N/A	N/A	N/A	N/A	N/A	1500 (1) 1650 (1)
Motor HP (Qty)	N/A	N/A	N/A	N/A	N/A	N/A	3.85 (1) 3.90 (1)

SCS-MC Air Cooled Condensers - Single Circuit							
SCS-MC	**015**	**018**	**031**	**035**	**056**	**071**	**111**
-SEC Intelligent Control Package with EC Fans							
208–230/1/60, 277/1/60							
Fan RPM	1630	1630	N/A	N/A	N/A	N/A	N/A
Motor HP (Qty)	0.55 (1)	0.55 (1)	N/A	N/A	N/A	N/A	N/A
208–230/3/60							
Fan RPM	N/A	N/A	1450	1450	1500	1500	1500
Motor HP (Qty)	N/A	N/A	1.39 (1)	1.39 (1)	3.85 (1)	3.85 (1)	3.85 (2)
460/3/60, 575/3/60							
Fan RPM	N/A	N/A	1440	1440	1500	1500	1500
Motor HP (Qty)	N/A	N/A	1.41 (1)	1.41 (1)	3.85 (1)	3.85 (1)	3.85 (2)
-SLN Low Noise Intelligent Control Package with EC Fans							
208–230/1/60, 277/1/60							
Fan RPM	1000	1000	N/A	N/A	N/A	N/A	N/A
Motor HP (Qty)	0.55 (1)	0.55 (1)	N/A	N/A	N/A	N/A	N/A
208–230/3/60							
Fan RPM	N/A	N/A	890	890	915	915	915
Motor HP (Qty)	N/A	N/A	1.39 (1)	1.39 (1)	3.85 (1)	3.85 (1)	3.85 (2)
460/3/60, 575/3/60							
Fan RPM	N/A	N/A	885	885	915	915	915
Motor HP (Qty)	N/A	N/A	1.41 (1)	1.41 (1)	3.85 (1)	3.85 (1)	3.85 (2)

SCS-MC Air Cooled Condensers - Dual Circuit										
SCS-MC	**031**	**035**	**056**	**071**	**111**	**128**	**142**	**223**	**264**	**334**
-DAA Fan Cycling Control Package										
208–230/1/60, 277/1/60										
Fan RPM	1490	1490	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Motor HP (Qty)	1.23 (1)	1.23 (1)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
208–230/3/60										
Fan RPM	1445	1445	1585	1585	1585	1585	1585	1585	1585	1585
Motor HP (Qty)	1.58 (1)	1.58 (1)	3.58 (1)	3.58 (1)	3.58 (2)	3.58 (2)	3.58 (2)	3.58 (4)	3.58 (4)	3.58 (6)
460/3/60, 575/3/60										
Fan RPM	1545	1545	1650	1650	1650	1650	1650	1650	1650	1650
Motor HP (Qty)	1.76 (1)	1.76 (1)	3.90 (1)	3.90 (1)	3.90 (2)	3.90 (2)	3.90 (2)	3.90 (4)	3.90 (4)	3.90 (6)
-DSA Variable Fan Speed Control Package										
208–230/1/60, 277/1/60										
Fan RPM	1490	1490	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Motor HP (Qty)	1.23 (1)	1.23 (1)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
208–230/3/60										
Fan RPM	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1500 (2) 1585 (2)	1500 (2) 1585 (2)	1500 (2) 1585 (4)
Motor HP (Qty)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.85 (2) 3.58 (2)	3.85 (2) 3.58 (2)	3.85 (2) 3.58 (4)
460/3/60, 575/3/60										
Fan RPM	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1500 (2) 1650 (2)	1500 (2) 1650 (2)	1500 (2) 1650 (4)
Motor HP (Qty)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.85 (2) 3.90 (2)	3.85 (2) 3.90 (2)	3.85 (2) 3.90 (4)

SCS-MC Air Cooled Condensers - Dual Circuit										
SCS-MC	**031**	**035**	**056**	**071**	**111**	**128**	**142**	**223**	**264**	**334**
-DEC Intelligent Control Package with EC Fans										
208–230/3/60										
Fan RPM	1450	1450	1500	1500	1500	1500	1500	1500	1500	1500
Motor HP (Qty)	1.39 (1)	1.39 (1)	3.85 (1)	3.85 (1)	3.85 (2)	3.85 (2)	3.85 (2)	3.85 (4)	3.85 (4)	3.85 (6)
460/3/60, 575/3/60										
Fan RPM	1440	1440	1500	1500	1500	1500	1500	1500	1500	1500
Motor HP (Qty)	1.41 (1)	1.41 (1)	3.85 (1)	3.85 (1)	3.85 (2)	3.85 (2)	3.85 (2)	3.85 (4)	3.85 (4)	3.85 (6)
-DLN Low Noise Intelligent Control Package with EC Fans										
208–230/3/60										
Fan RPM	890	890	915	915	915	915	915	915	915	915
Motor HP (Qty)	1.39 (1)	1.39 (1)	3.85 (1)	3.85 (1)	3.85 (2)	3.85 (2)	3.85 (2)	3.85 (4)	3.85 (4)	3.85 (6)
460/3/60, 575/3/60										
Fan RPM	885	885	915	915	915	915	915	915	915	915
Motor HP (Qty)	1.41 (1)	1.41 (1)	3.85 (1)	3.85 (1)	3.85 (2)	3.85 (2)	3.85 (2)	3.85 (4)	3.85 (4)	3.85 (6)

4. Electrical Data

4.1. No Receiver Skid, Single and Dual Circuit

4.1.1. AA Models

Model (No Receiver)	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-MC-015-( )AA	2.0	15	1.8	15	N/A		0.4	15	0.3	15
SCS-MC-018-( )AA	2.0	15	1.8	15	N/A		0.4	15	0.3	15
SCS-MC-031-( )AA	5.4	15	4.9	15	3.8	15	2.0	15	1.6	15
SCS-MC-035-( )AA	5.4	15	4.9	15	3.8	15	2.0	15	1.6	15
SCS-MC-056-( )AA	N/A		N/A		11.4	20	6.2	15	4.9	15
SCS-MC-071-( )AA	N/A		N/A		11.4	20	6.2	15	4.9	15
SCS-MC-111-( )AA	N/A		N/A		22.7	25	12.3	15	9.9	15
SCS-MC-128-( )AA	N/A		N/A		22.7	25	12.3	15	9.9	15
SCS-MC-142-( )AA	N/A		N/A		22.7	25	12.3	15	9.9	15
SCS-MC-223-( )AA	N/A		N/A		45.4	50	24.7	25	19.7	20
SCS-MC-264-( )AA	N/A		N/A		45.4	50	24.7	25	19.7	20
SCS-MC-334-( )AA	N/A		N/A		68.2	70	37.0	40	29.6	30

Key:
MCA = Minimum Circuit Amps
MFS = Maximum Fuse Size
FLA = Full Load Amps, listed per component on unit nameplates
Format = Voltage/Number of Phases/Frequency. E.g., 277/1/60 = 277 V / 1 phase / 60 Hz

4.1.2. SA Models

Model (No Receiver)	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-MC-015-( )SA	3.6	15	2.7	15	N/A		N/A		N/A	
SCS-MC-018-( )SA	3.6	15	2.7	15	N/A		N/A		N/A	
SCS-MC-031-( )SA	5.4	15	4.9	15	N/A		N/A		N/A	
SCS-MC-035-( )SA	5.4	15	4.9	15	N/A		N/A		N/A	
SCS-MC-111-( )SA	N/A		N/A		24.5	30	12.2	15	9.7	15
SCS-MC-223-( )SA	N/A		N/A		47.2	50	24.5	25	19.6	20
SCS-MC-264-( )SA	N/A		N/A		47.2	50	24.5	25	19.6	20
SCS-MC-334-( )SA	N/A		N/A		70.0	70	36.9	40	29.5	30

4.1.3. EC and LN Models

Model (No Receiver)	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-MC-015-( )EC/LN	3.6	15	2.7	15	N/A		N/A		N/A	
SCS-MC-018-( )EC/LN	3.6	15	2.7	15	N/A		N/A		N/A	
SCS-MC-031-( )EC/LN	N/A		N/A		4.0	15	2.9	15	2.3	15
SCS-MC-035-( )EC/LN	N/A		N/A		4.0	15	2.9	15	2.3	15
SCS-MC-056-( )EC/LN	N/A		N/A		12.3	20	6.1	15	4.9	15
SCS-MC-071-( )EC/LN	N/A		N/A		12.3	20	6.1	15	4.9	15
SCS-MC-111-( )EC/LN	N/A		N/A		24.5	30	12.2	15	9.7	15
SCS-MC-128-( )EC/LN	N/A		N/A		24.5	30	12.2	15	9.7	15
SCS-MC-142-( )EC/LN	N/A		N/A		24.5	30	12.2	15	9.7	15
SCS-MC-223-( )EC/LN	N/A		N/A		49.0	50	24.3	25	19.5	20
SCS-MC-264-( )EC/LN	N/A		N/A		49.0	50	24.3	25	19.5	20
SCS-MC-334-( )EC/LN	N/A		N/A		73.6	80	36.5	40	29.2	30

4.2. With Receiver Skid, Single Circuit

4.1.4. SAA Models

Model (Single with Receiver)	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-MC-015-SAA	2.7	15	2.4	15	N/A		0.8	15	0.6	15
SCS-MC-018-SAA	2.7	15	2.4	15	N/A		0.8	15	0.6	15
SCS-MC-031-SAA	6.1	15	5.4	15	4.6	15	2.4	15	1.9	15
SCS-MC-035-SAA	6.1	15	5.4	15	4.6	15	2.4	15	1.9	15
SCS-MC-056-SAA	N/A		N/A		12.1	20	6.5	15	5.2	15
SCS-MC-071-SAA	N/A		N/A		12.1	20	6.5	15	5.2	15
SCS-MC-111-SAA	N/A		N/A		23.5	30	12.7	15	10.1	15

4.2.1. SSA Models

Model (Single with Receiver)	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-MC-015-SSA	4.3	15	3.3	15	N/A		N/A		N/A	
SCS-MC-018-SSA	4.3	15	3.3	15	N/A		N/A		N/A	
SCS-MC-031-SSA	6.1	15	5.4	15	N/A		N/A		N/A	
SCS-MC-035-SSA	6.1	15	5.4	15	N/A		N/A		N/A	
SCS-MC-111-SSA	N/A		N/A		24.4	30	12.6	15	10.1	15

4.2.2. SEC and SLN Models

Model (Single with Receiver)	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-MC-015-SEC/SLN	4.3	15	3.3	15	N/A		N/A		N/A	
SCS-MC-018-SEC/SLN	4.3	15	3.3	15	N/A		N/A		N/A	
SCS-MC-031-SEC/SLN	N/A		N/A		4.8	15	3.2	15	3.2	15
SCS-MC-035-SEC/SLN	N/A		N/A		4.8	15	3.2	15	3.2	15
SCS-MC-056-SEC/SLN	N/A		N/A		13.0	20	6.4	15	6.4	15
SCS-MC-071-SEC/SLN	N/A		N/A		13.0	20	6.4	15	6.4	15
SCS-MC-111-SEC/SLN	N/A		N/A		25.3	30	12.5	15	12.5	15

4.3. With Receiver Skid, Dual Circuit

4.2.3. DAA Models

Model (Dual with Receiver)	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-MC-031-DAA	6.9	15	6.0	15	5.3	15	2.7	15	2.2	15
SCS-MC-035-DAA	6.9	15	6.0	15	5.3	15	2.7	15	2.2	15
SCS-MC-056-DAA	N/A		N/A		12.9	20	6.8	15	5.5	15
SCS-MC-071-DAA	N/A		N/A		12.9	20	6.8	15	5.5	15
SCS-MC-111-DAA	N/A		N/A		24.2	30	13.0	15	10.4	15
SCS-MC-128-DAA	N/A		N/A		24.2	30	13.0	15	10.4	15
SCS-MC-142-DAA	N/A		N/A		24.2	30	13.0	15	10.4	15
SCS-MC-223-DAA	N/A		N/A		46.9	50	25.4	30	20.3	25
SCS-MC-264-DAA	N/A		N/A		46.9	50	25.4	30	20.3	25
SCS-MC-334-DAA	N/A		N/A		68.9	70	37.4	40	29.9	30

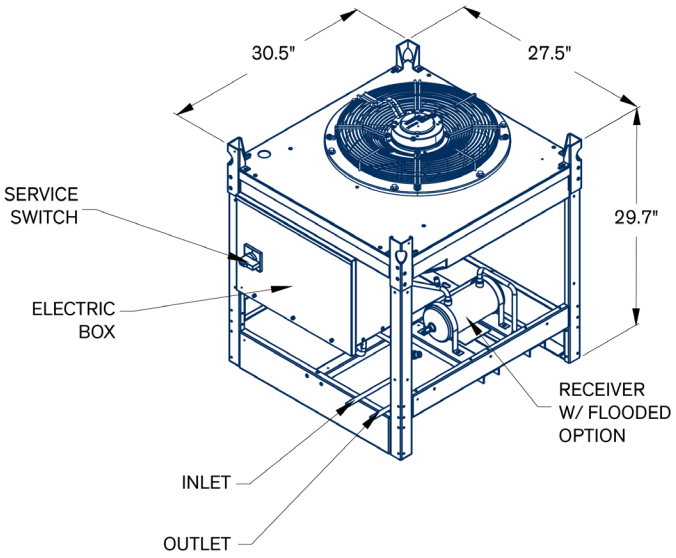
4.3.1. DSA Models

Model (Dual with Receiver)	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-MC-031-DSA	6.9	15	6.0	15	N/A		N/A		N/A	
SCS-MC-035-DSA	6.9	15	6.0	15	N/A		N/A		N/A	
SCS-MC-223-DSA	N/A		N/A		47.2	50	24.5	25	19.6	20
SCS-MC-264-DSA	N/A		N/A		47.2	50	24.5	25	19.6	20
SCS-MC-334-DSA	N/A		N/A		70.7	80	37.2	40	29.8	30

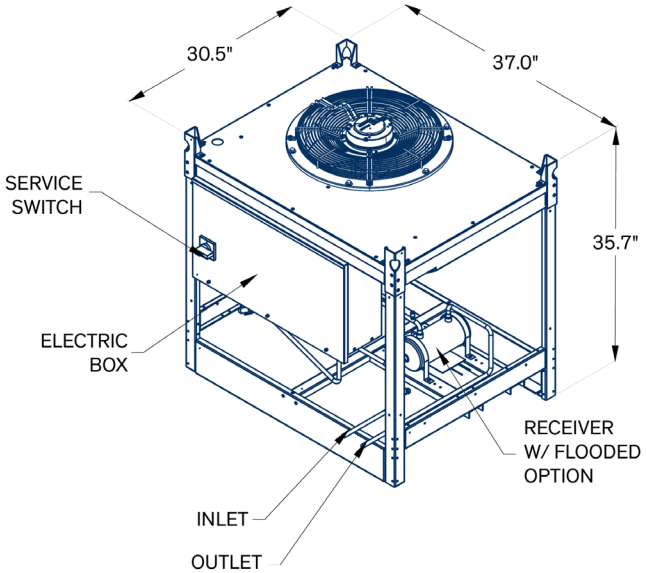
4.3.2. DEC and DLN Models

Model (Dual with Receiver)	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-MC-031-DEC/DLN	N/A		N/A		5.5	15	3.6	15	2.9	15
SCS-MC-035-DEC/DLN	N/A		N/A		5.5	15	3.6	15	2.9	15
SCS-MC-056-DEC/DLN	N/A		N/A		13.8	20	6.8	15	5.4	15
SCS-MC-071-DEC/DLN	N/A		N/A		13.8	20	6.8	15	5.4	15
SCS-MC-111-DEC/DLN	N/A		N/A		26.0	30	12.8	15	10.3	15
SCS-MC-128-DEC/DLN	N/A		N/A		26.0	30	12.8	15	10.3	15
SCS-MC-142-DEC/DLN	N/A		N/A		26.0	30	12.8	15	10.3	15
SCS-MC-223-DEC/DLN	N/A		N/A		50.5	60	25.0	30	20.0	25
SCS-MC-264-DEC/DLN	N/A		N/A		50.5	60	25.0	30	20.0	25
SCS-MC-334-DEC/DLN	N/A		N/A		74.3	80	36.9	40	29.5	30

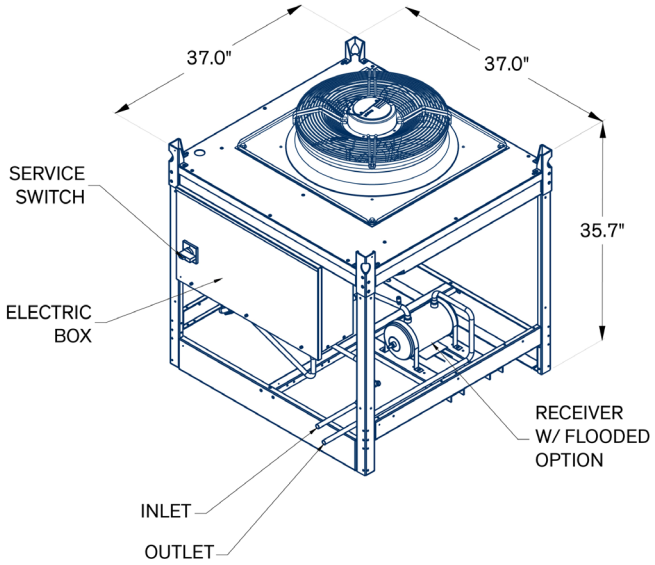
5. Dimensional Data



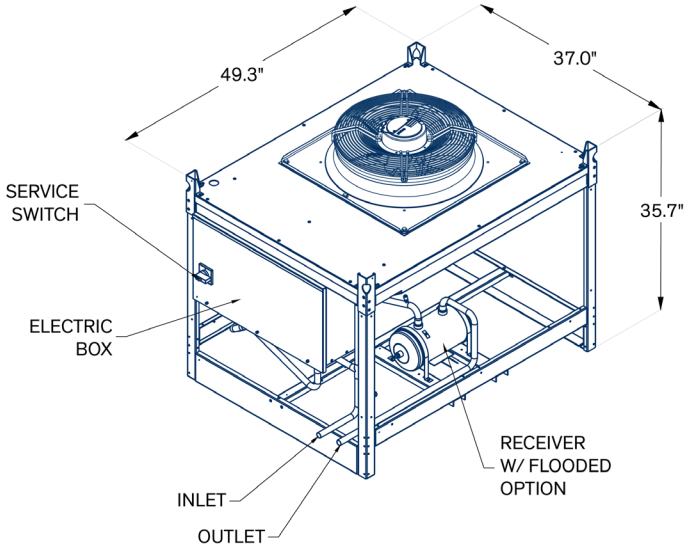
4.4. SCS-MC-015-S



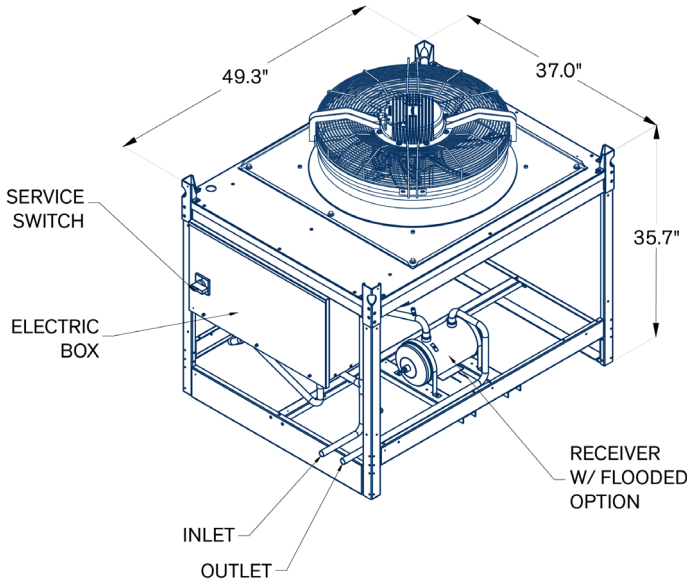
5.1. SCS-MC-018-S



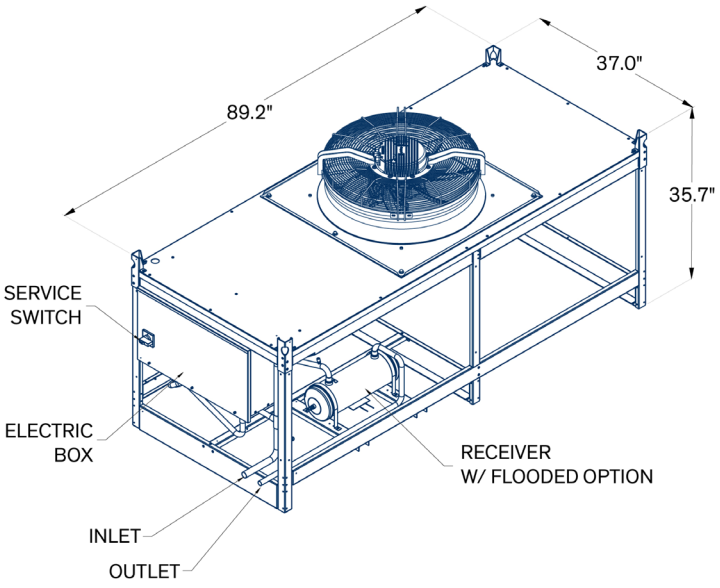
5.2. SCS-MC-031-S



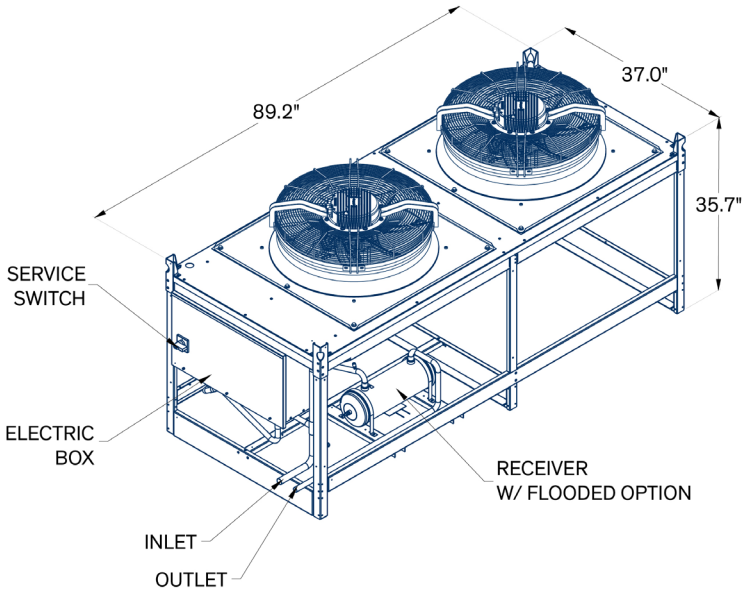
5.3. SCS-MC-035-S



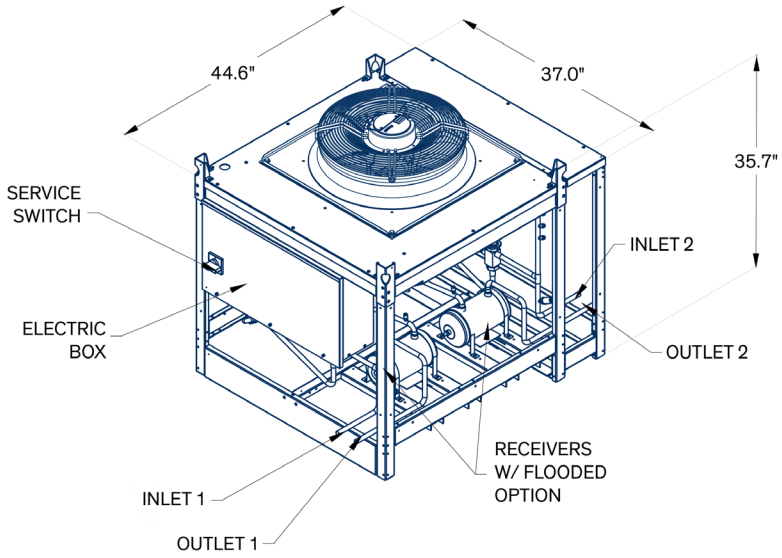
5.4. SCS-MC-056-S



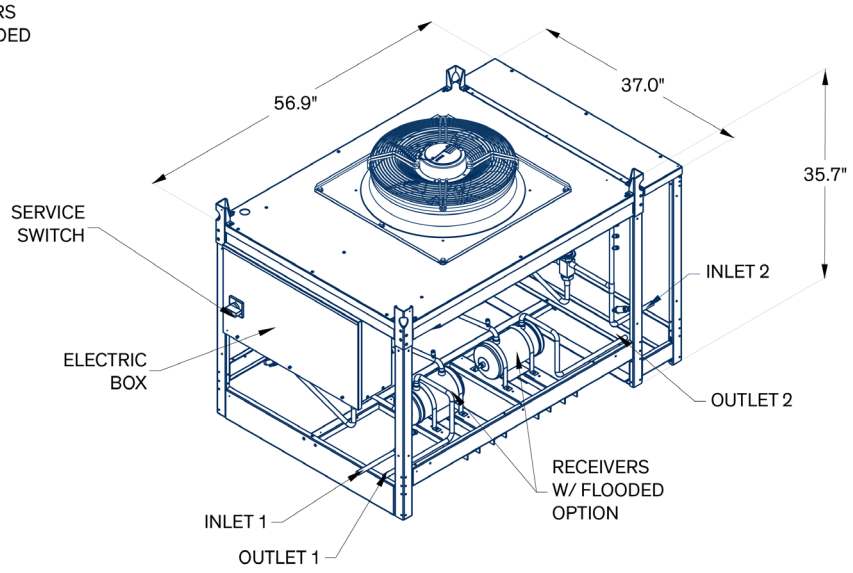
5.5. SCS-MC-071-S



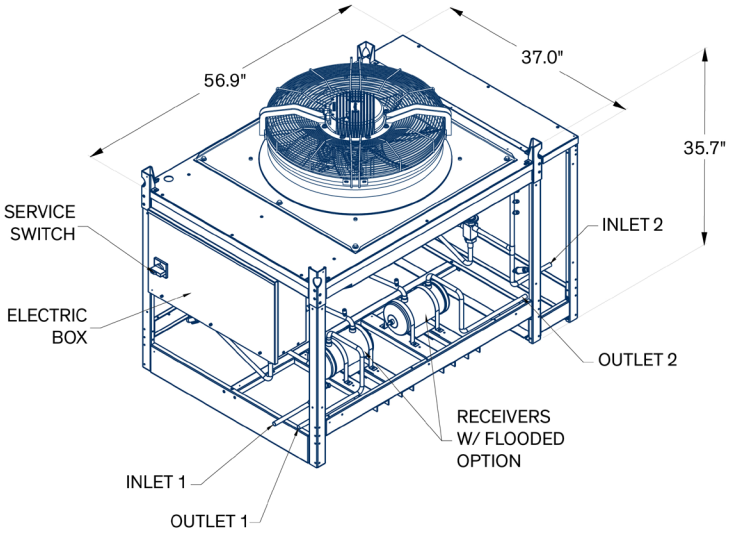
5.6. SCS-MC-111-S



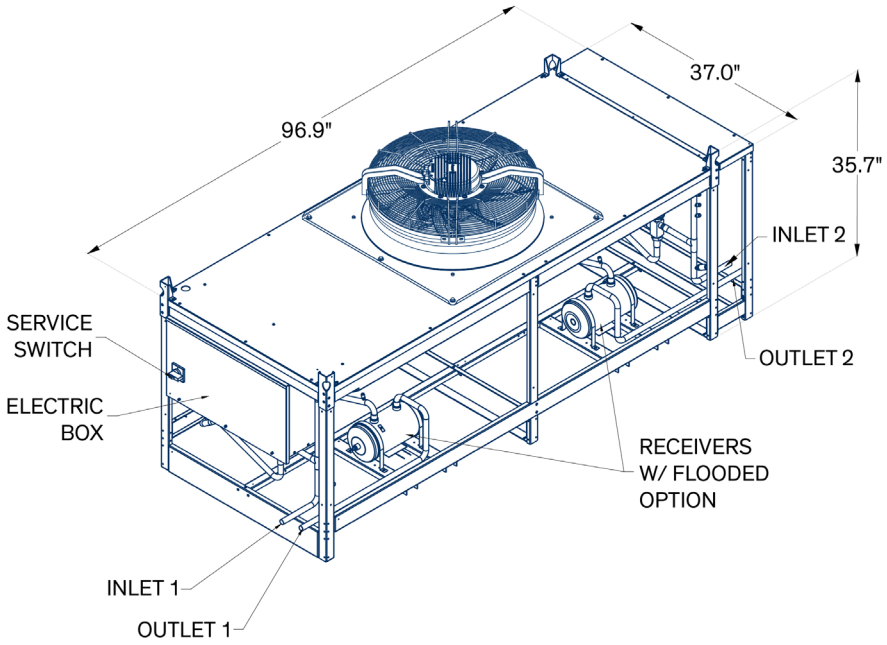
5.7. SCS-MC-031-D



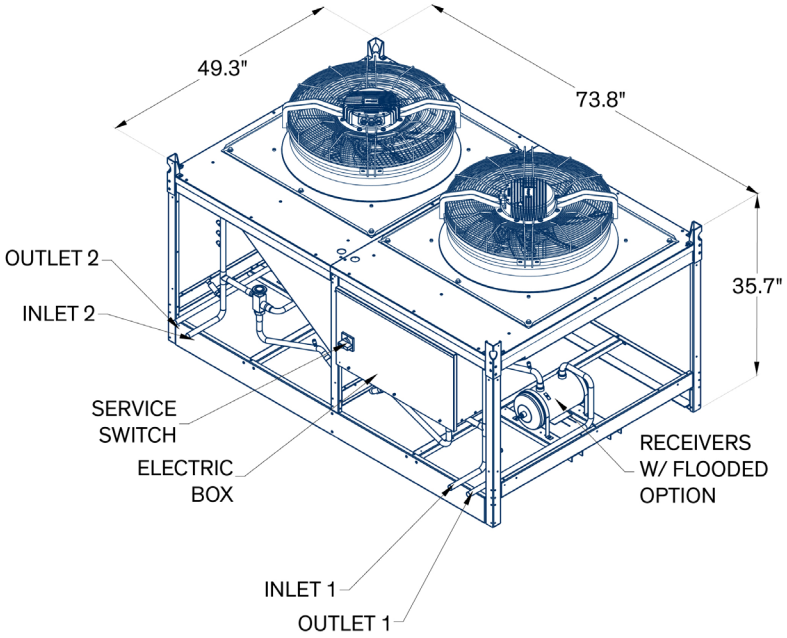
5.8. SCS-MC-035-D



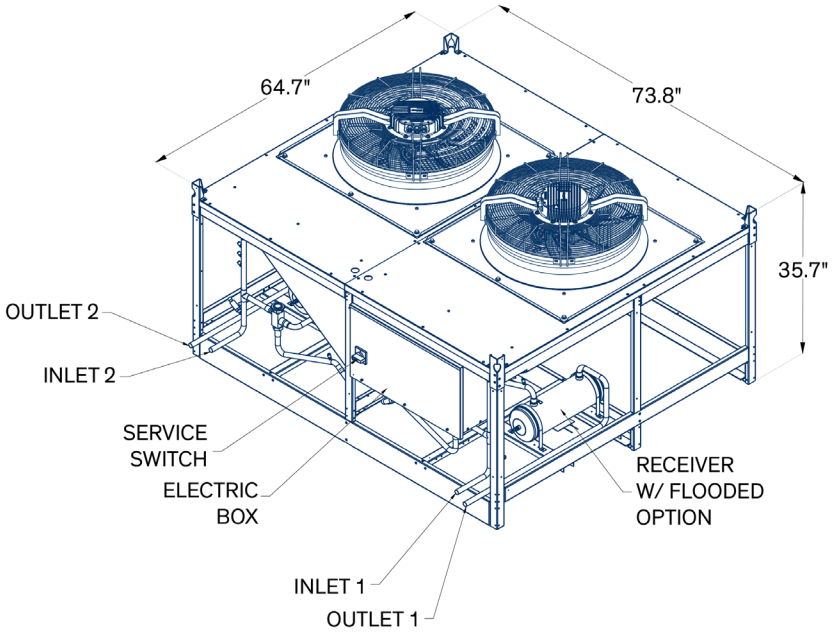
5.9. SCS-MC-056-D



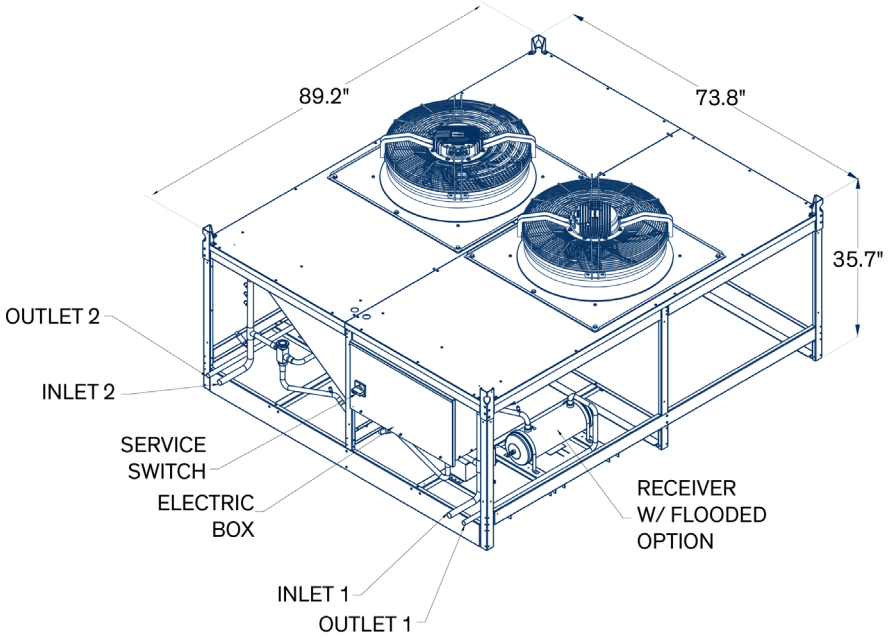
5.10. SCS-MC-071-D



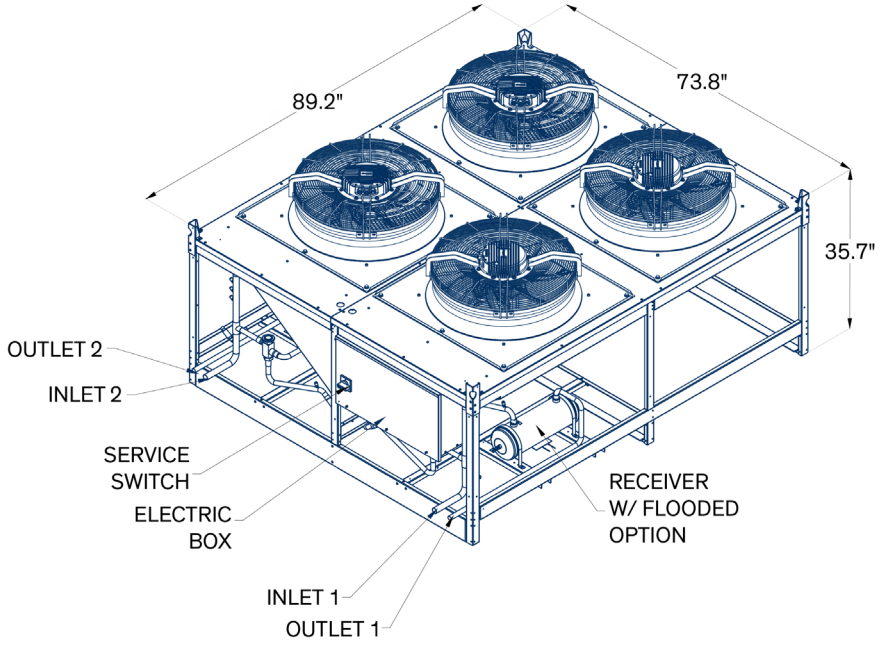
5.11. SCS-MC-111-D



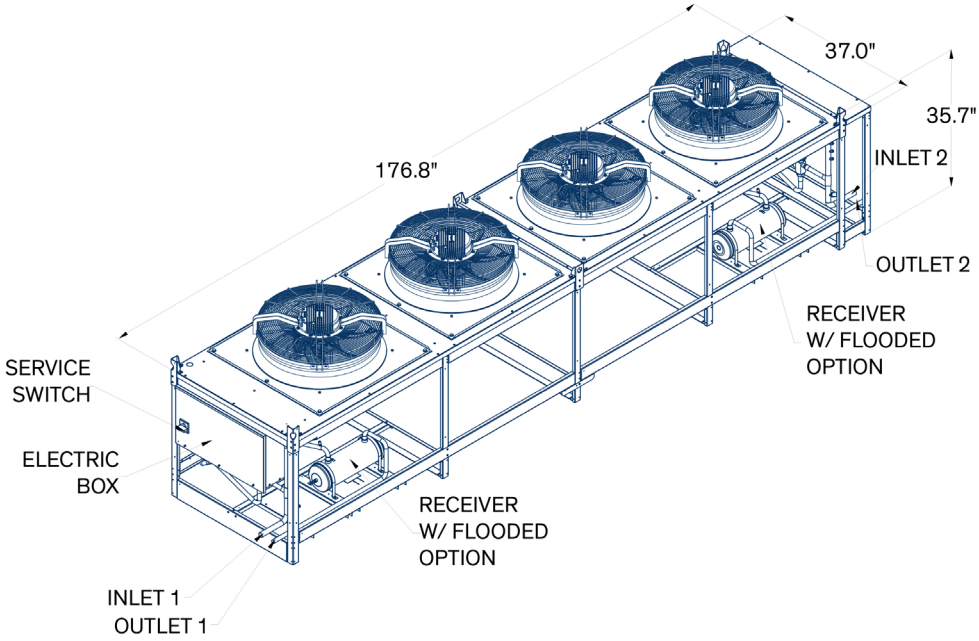
5.12. SCS-MC-128-D



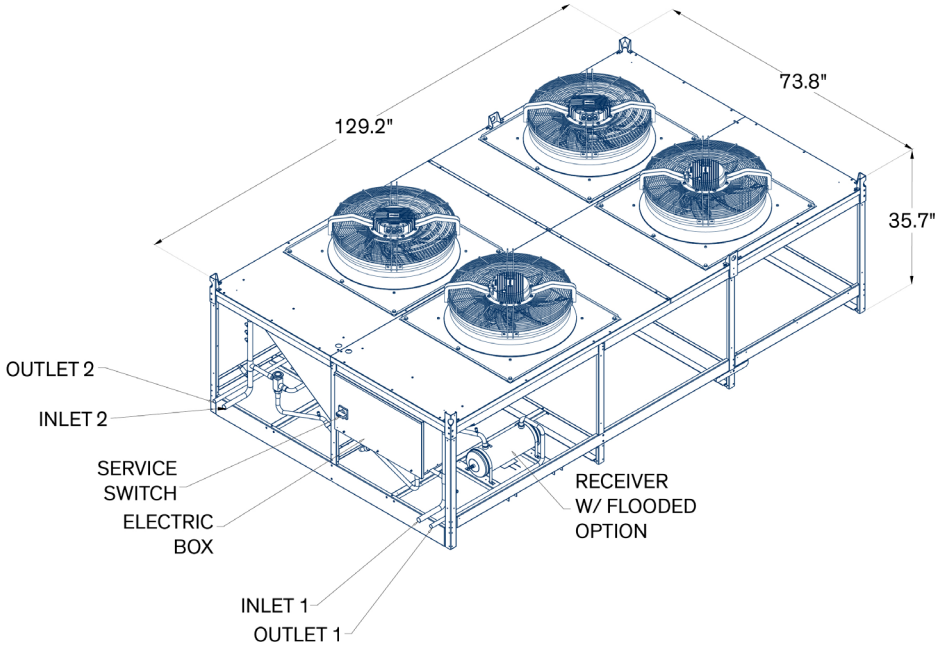
5.13. SCS-MC-142-D



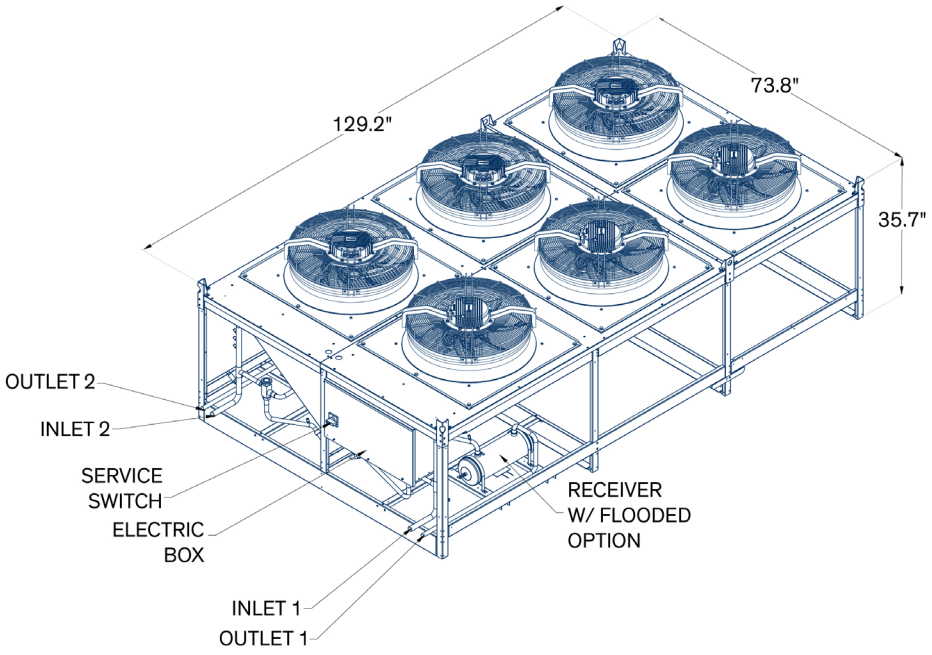
5.14. SCS-MC-223-D (2×2)



5.15. SCS-MC-223-D (1×4)



5.16. SCS-MC-264-D



5.17. SCS-MC-334-D

6. Guide Specifications

SCS-MC-XXX-XXX-XX, Micro-Channel Air-Cooled Condenser

6.1. Summary

This specification describes requirements for a refrigerant condensing system to be used with a DX-based precision environmental control system. The STULZ Micro-Channel Condenser is an outdoor air-cooled condenser with a direct driven, external rotor-motor integrated in axial fans. The Micro-Channel Condenser is a high efficiency state-of-the-art condenser that provides a high total heat of rejection with reduced weight and a smaller footprint than other condensers. The Micro-Channel Condenser is designed for single or dual refrigeration circuits. The Micro-Channel Condenser model number shall be SCS-MC-XXX-XXX-XX.

6.1.1. Design Requirements

The units are designed for outdoor installation with a removable front electric access panel. They do not require side access, however the side and rear are accessible.

6.1.2. Refrigerant

Condensing units are designed for use with R407C or R410A refrigerant. The system is provided with a dry nitrogen charge and requires field evacuation and refrigerant charging.

6.1.3. Quality Assurance

STULZ maintains a set of international standards of quality management to ensure product quality. STULZ is ISO 9001 certified. Before shipment, each system is tested for complete operation and function.

6.2. Cabinet

The condenser cabinets are constructed of 0.090 in aluminum and are securely fastened to a frame constructed of 0.125 in aluminum. The condenser cabinet houses the condenser coil, fans, fan guards, condenser motor control, and NEMA 3R electric box. Optionally, a receiver is piped and mounted to the cabinet frame.

6.3. Head Pressure Control

Four options, must choose one: 0°F fan cycling, -20°F variable fan speed control, -20°F intelligent control, or -20°F low noise intelligent control. And as an add on: -30°F flooded control with receiver.

All head pressure control options include refrigerant pressure transducers that monitor refrigerant discharge pressure then adjusts fans in response. Increased air flow over condenser coils enhances heat rejection, causing head pressure to decrease as cooling improves, and vice versa.

6.3.1. (AA) 0°F Fan Cycling Control (OHS only)

The unit cycles AC fans for low-ambient cooling control. This control maintains year-round air conditioning at a minimum ambient air temperature of 0°F DB. By cycling the condenser fans on refrigerant pressure, the refrigerant head pressure is maintained.

6.3.2. (SA) -20°F Variable Fan Speed Control

The unit blows a variable-speed fan for low-ambient cooling control, and cycles remaining fans as needed. This control

maintains year-round air conditioning at a minimum ambient air temperature of -20°F DB. The speed of EC fan number one (closest to the header) is varied based on refrigerant discharge pressure. All remaining fans shall be controlled by fan cycling and are AC.

6.3.3. (EC) -20°F Intelligent Control

The unit blows variable-speed EC fans for low-ambient cooling control. This control maintains year-round air conditioning at a minimum ambient air temperature of -20°F DB. All fans are highly efficient, electronically commutated axial fans. All the EC fans are precisely controlled with variable speeds per pressure transducers to maintain refrigerant discharge pressure.

6.3.4. (LN) -20°F Low Noise Intelligent Control

The unit blows variable-speed EC fans for low-ambient cooling control. This control maintains year-round air conditioning at a minimum ambient air temperature of -20°F DB. Low noise EC fans operate at a lower RPM. All fans are low noise, highly efficient, electronically commutated axial fans. The low noise fans shall meet the customer requirements and required maximum noise emissions standards. All the low noise EC fans are precisely controlled with variable speeds per pressure transducers to maintain refrigerant discharge pressure.

6.3.5. (F) -30°F Flooded Control with Receiver

The -30°F flooded control with receiver option is available in combination with any other head pressure control.

The unit runs a low-ambient flooded-head pressure control via a receiver and valve. This pressure control maintains year-round air conditioning at a minimum ambient air temperature of -30°F DB. This option includes an insulated liquid refrigerant receiver with a liquid sight glass, head pressure control valves (for flooded condenser operation), and a receiver heater pad with internal thermostat control. The head pressure control valve and receiver heater pad are factory installed. The condenser includes branch circuit protection and power source for the receiver heater pads.

6.4. Mechanical Components

6.4.1. AC and EC Axial Fans

The fans shall be direct-driven, external rotor-motor integrated in an axial fan unit. The fan blades shall be constructed of a weather resistant, long-lasting coated steel or aluminum. Each fan is low noise and low vibration. Each fan impeller is dynamically and statically balanced in two planes to minimize vibration during operation.

6.4.2. Micro-Channel Coil

The micro-channel coil is constructed of brazed aluminum. The coil is designed with high performance fins to provide low airside pressure drop and high heat transfer. Micro-channel tubes offer more predictable performance and achieve improved air-to-refrigerant approach temperatures.

Optional coil coating shall be available.

6.5. Electrical System

The electrical system conforms to National Electrical Code (NEC) class II requirements. The control circuit is 24 VAC. The control circuit wire is not smaller than 18 AWG. All wiring is neatly wrapped, run in conduit or cable trays, and routed in bundles. Each wire ends with a service loop and is securely fastened per the NEC. Each wire is numbered for ease of service tracing.

All electrically actuated components are safely accessible from the unit's front without reaching over exposed high-voltage or rotating parts. Circuit breakers or manual motor starters protect each high-voltage circuit in all three phases. The blower

motors have thermal and short circuit protection. Wiring is routed in separate bundles for line voltage and 24 V control circuits.

The electric box is conveniently located for full-service access. It includes all components required for unit operation, including contactors, starters, fuses, circuit breakers, and terminal boards.

6.5.1. Main Power Service Switch

The micro-channel condenser is mounted with a main power service switch.

6.6. Code Conformance

The supplied SCS Micro-Channel Condenser system meets the following compliance approvals:

- c**ETL**us listed and conforms to UL standard 1995 (2015 Ed. 5)
- Certified to CSA standard C22.2 # 236 (2015 Ed. 5)
- Intertek control number 55789

SCS Outdoor Air-Cooled Condensers

Model SCS-940-D Shown

See model nomenclature on next page.

7. Nomenclature

SCS-XXX-XXX			
SCS	**Capacity (MBH or 1,000s of Btu/h)**	**Number of Coolant Circuits**	**Fan Options**
SCS = Outdoor Air-Cooled Condenser	012 018 024 036	S = Single	AA = Fan Cycling Control, AC Fans SA = Variable Speed Control, EC and AC Fans EC = Intelligent Control, Electronically Commutated Fans
	060 096 120 144 192 252	S = Single D = Dual	
	276 312 447 525 597 683 940 1366	D = Dual	

Features

- R410A or R407C refrigerant
- All aluminum cabinet construction
- Uses highly efficient axial fans with AC and EC motor configurations
- Integrated control box with main disconnect and motor protection
- Various fan speed control options based on refrigerant pressure
- Single and dual circuit configurations
- Low ambient temperature control options (-30°F)
- Receiver skid provided for -30°F applications

8. Performance Data: AA, SA, & EC Models

Model	Flow Rate	Total Heat Rejection w/ R410A (MBH)		Total Heat Rejection w/ R407C (MBH)		Noise	Refrigerant Connections (in)	
	(CFM)	@ 95°F	@ 105°F	@ 95°F	@ 105°F	(dBA)	Hot Gas Line	Liquid Line
SCS-012-S	2,575	17	N/A	18	N/A	73	5/8	5/8
SCS-018-S	2,450	22	N/A	25	N/A	73	5/8	5/8
SCS-024-S	2,350	31	20	35	22	73	7/8	7/8
SCS-036-S	4,900	51	32	57	35	79	7/8	7/8
SCS-060-S	4,550	66	41	75	45	79	7/8	7/8
SCS-060-D	4,550	66	41	75	45	79	7/8	7/8
SCS-096-S	9,775	102	63	115	71	82	1 1/8	7/8
SCS-096-D	9,775	102	63	115	71	82	1 1/8	7/8
SCS-120-S	9,100	135	86	148	94	82	1 1/8	7/8
SCS-120-D	9,100	135	86	148	94	82	1 1/8	7/8
SCS-144-S	8,675	152	97	170	104	82	1 1/8	7/8
SCS-144-D	8,675	152	97	170	104	82	1 1/8	7/8
SCS-192-S	19,325	246	157	243	154	85	1 1/8	7/8
SCS-192-D	19,325	246	157	243	154	85	1 1/8	7/8
SCS-252-S	18,400	331	208	326	204	85	1 3/8	1 1/8
SCS-252-D	18,409	331	208	326	204	85	1 3/8	1 1/8
SCS-276-D	18,475	352	221	355	213	85	1 3/8	1 1/8
SCS-312-D	17,525	400	252	396	248	85	1 3/8	1 1/8
SCS-447-D	15,925	429	272	438	275	85	1 3/8	1 1/8
SCS-525-D	27,975	530	333	532	332	87	1 5/8	1 3/8
SCS-597-D	26,300	602	380	595	372	87	1 5/8	1 3/8
SCS-683-D	25,525	644	409	688	449	87	1 5/8	1 3/8
SCS-940-D	33,650	859	545	1032	598	88	2 1/8	1 5/8
SCS-1366-D	50,500	1,288	817	1376	897	90	2 1/8	1 5/8

8.1. Elevation Correction

Elevation above sea level has an effect on the performance of air-cooled condenser. Multiply the rated condenser's total heat rejection (THR) at sea level by the elevation correction factor in the table below when selecting the required condenser.

Elevation Correction Factors											
Elevation (ft)	1,000	2,000	3,000	4,000	5,000	6,000	8,000	10,000	12,000	14,000	16,000
Correction Factor	0.94	0.93	0.90	0.88	0.86	0.83	0.79	0.75	0.71	0.66	0.62

9. Technical Data

9.1. Dimensions and Weights

Model	Unit		Shipping	
	Overall Dimensions	Weight (lb)	Approximate Dimensions	Weight (lb)
SCS-012-S	27.5" W x 31.0" H x 36.0" D	75	36" W x 43" H x 48" D	100
SCS-018-S	27.5" W x 31.0" H x 36.0" D	80	36" W x 43" H x 48" D	105
SCS-024-S	27.5" W x 31.0" H x 36.0" D	85	36" W x 43" H x 48" D	110
SCS-036-S	32.8" W x 35.5" H x 44.5" D	140	41" W x 48" H x 57" D	180
SCS-060-S	32.8" W x 35.5" H x 44.5" D	150	41" W x 48" H x 57" D	195
SCS-060-D	32.8" W x 35.5" H x 44.5" D	150	41" W x 48" H x 57" D	195
SCS-096-S	32.8" W x 35.5" H x 76.0" D	220	41" W x 48" H x 88" D	285
SCS-096-D	32.8" W x 35.5" H x 76.0" D	220	41" W x 48" H x 88" D	285
SCS-120-S	32.8" W x 35.5" H x 76.0" D	240	41" W x 48" H x 88" D	310
SCS-120-D	32.8" W x 35.5" H x 76.0" D	240	41" W x 48" H x 88" D	310
SCS-144-S	32.8" W x 35.5" H x 76.0" D	255	41" W x 48" H x 88" D	330
SCS-144-D	32.8" W x 35.5" H x 76.0" D	255	41" W x 48" H x 88" D	330
SCS-192-S	48.3" W x 48.0" H x 108.5" D	455	57" W x 60" H x 121" D	590
SCS-192-D	48.3" W x 48.0" H x 108.5" D	455	57" W x 60" H x 121" D	590
SCS-252-S	48.3" W x 48.0" H x 108.5" D	495	57" W x 60" H x 121" D	645
SCS-252-D	48.3" W x 48.0" H x 108.5" D	495	57" W x 60" H x 121" D	645
SCS-276-D	48.3" W x 48.0" H x 108.5" D	520	57" W x 60" H x 121" D	675
SCS-312-D	48.3" W x 48.0" H x 108.5" D	555	57" W x 60" H x 121" D	720
SCS-447-D	48.3" W x 48.0" H x 108.5" D	620	57" W x 60" H x 121" D	810
SCS-525-D	48.3" W x 48.0" H x 155.5" D	835	57" W x 60" H x 168" D	1,085
SCS-597-D	48.3" W x 48.0" H x 155.5" D	890	57" W x 60" H x 168" D	1,160
SCS-683-D	48.3" W x 48.0" H x 155.5" D	1000	57" W x 60" H x 168" D	1,300
SCS-940-D (1X4)	48.3" W x 48.0" H x 204.0" D	1300	57" W x 60" H x 216" D	1,690
SCS-940-D (2X2)	93.3" W x 48.0" H x 108.5" D	1300	102" W x 60" H x 121" D	1,690
SCS-1366-D	93.7" W x 48.0" H x 155.5" D	1950	102" W x 60" H x 168" D	2,535

9.2. Compatible A/C Models

SCS Compatible A/C Models				
AA, SA & EC Models	R410A		R407C	
	@ 95°F	@ 105°F	@ 95°F	@ 105°F
SCS-012	OHS-012		OHS-012	
SCS-018			CCU/D-41 OHS-018	
SCS-024	OHS-018	OHS-012	CCU/D-61 COS-024 OHS-024	OHS-012
SCS-036	OHS-024 OHS-032 OHS-040	OHS-018 OHS-024	CCU/D-81 OHS-032 OHS-040	CCU/D-41 CCU/D-61 COS-024 OHS-018 OHS-024
SCS-060	CRS-042	OHS-032	CCU/D-121 CCU/D-171 OHS-048	CCU/D-081 OHS-032 OHS-040
SCS-096	OHS-048 OHS-060	OHS-040	CCU/D-201 OHS-060 OHS-084	CCU/D-121 OHS-048
SCS-120	CFU/D-021 CRS-084 OHS-084	CRS-042 OHS-048 OHS-060	COS-060 COS-096	CCU/D-171 CCU/D-201 COS-060 OHS-060
SCS-144	CFU/D-028		COS-120	
SCS-192	CFU/D-035	CFU/D-021 CFU/D-028 OHS-084 CRS-084		COS-096 OHS-084
SCS-252	CFU/D-053 CFU/D-070	CFU/D-035		COS-120
SCS-276				
SCS-312	CFU/D-088			
SCS-447		CFU/D-053		
SCS-525	CFU/D-105	CFU/D-070		
SCS-597				
SCS-683		CFU/D-088		
SCS-940		CFU/D-105		
SCS-1366				

9.4. Fan and Motor Data, Single Circuit

Air Cooled Condensers, Single Circuit										
SCS	**012**	**018**	**024**	**036**	**060**	**096**	**120**	**144**	**192**	**252**
-SAA Fan Cycling Control Package										
208–230/1/60, 277/1/60										
Fan RPM	1260	1260	1260	1490	1490	1490	1490	1490	N/A	N/A
Motor HP (Qty)	0.35 (1)	0.35 (1)	0.35 (1)	1.23 (1)	1.23 (1)	1.23 (2)	1.23 (2)	1.23 (2)	N/A	N/A
208–230/3/60										
Fan RPM	N/A	N/A	N/A	1445	1445	1445	1445	1445	1585	1586
Motor HP (Qty)	N/A	N/A	N/A	1.58 (1)	1.58 (1)	1.58 (2)	1.58 (2)	1.58 (2)	3.58 (2)	3.58 (2)
460/3/60, 575/3/60										
Fan RPM	1520	1520	1520	1545	1545	1545	1545	1545	1650	1651
Motor HP (Qty)	0.31 (1)	0.31 (1)	0.31 (1)	1.76 (1)	1.76 (1)	1.76 (2)	1.76 (2)	1.76 (2)	3.90 (2)	3.90 (2)
-SSA Variable Fan Speed Control Package										
208–230/1/60, 277/1/60										
Fan RPM	1260	1260	1260	1490	1490	1490	1490	1490	N/A	N/A
Motor HP (Qty)	0.35 (1)	0.35 (1)	0.35 (1)	1.23 (1)	1.23 (1)	1.23 (2)	1.23 (2)	1.23 (2)	N/A	N/A
208–230/3/60										
Fan RPM	N/A	N/A	N/A	N/A	N/A	1490 (1) 1445 (1)	1490 (1) 1445 (1)	1490 (1) 1445 (1)	1500 (1) 1585 (1)	1500 (1) 1585 (1)
Motor HP (Qty)	N/A	N/A	N/A	N/A	N/A	1.23 (1) 1.58 (1)	1.23 (1) 1.58 (1)	1.23 (1) 1.58 (1)	3.85 (1) 3.58 (1)	3.85 (1) 3.58 (1)
460/3/60, 575/3/60										
Fan RPM	N/A	N/A	N/A	N/A	N/A	1440 (1) 1545 (1)	1440 (1) 1545 (1)	1440 (1) 1545 (1)	1500 (1) 1650 (1)	1500 (1) 1650 (1)
Motor HP (Qty)	N/A	N/A	N/A	N/A	N/A	1.41 (1) 1.76 (1)	1.41 (1) 1.76 (1)	1.41 (1) 1.76 (1)	3.85 (1) 3.90 (1)	3.85 (1) 3.90 (1)
-SEC Intelligent Control Package with EC Fans										
208–230/1/60, 277/1/60										
Fan RPM	1630	1630	1630	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Motor HP (Qty)	0.55 (1)	0.55 (1)	0.55 (1)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
208–230/3/60										
Fan RPM	N/A	N/A	N/A	1450	1450	1450	1450	1450	1500	1500
Motor HP (Qty)	N/A	N/A	N/A	1.39 (1)	1.39 (1)	1.39 (2)	1.39 (2)	1.39 (2)	3.85 (2)	3.85 (2)
460/3/60, 575/3/60										
Fan RPM	N/A	N/A	N/A	1440	1440	1440	1440	1440	1500	1500
Motor HP (Qty)	N/A	N/A	N/A	1.41 (1)	1.41 (1)	1.41 (2)	1.41 (2)	1.41 (2)	3.85 (2)	3.85 (2)

9.3. Fan and Motor Data, Dual Circuit

SCS Air Cooled Condensers, Dual Circuit							
SCS	**060**	**096**	**120**	**144**	**192**	**252**	**276**
-DAA Fan Cycling Control Package							
208–230/1/60, 277/1/60							
Fan RPM	1490	1490	1490	1490	N/A	N/A	N/A
Motor HP (Qty)	1.23 (1)	1.23 (2)	1.23 (2)	1.23 (2)	N/A	N/A	N/A
208–230/3/60							
Fan RPM	1445	1445	1445	1445	1585	1585	1585
Motor HP (Qty)	1.58 (1)	1.58 (2)	1.58 (2)	1.58 (2)	3.58 (2)	3.58 (2)	3.58 (2)
460/3/60, 575/3/60							
Fan RPM	1545	1545	1545	1545	1650	1650	1650
Motor HP (Qty)	1.76 (1)	1.76 (2)	1.76 (2)	1.76 (1)	3.90 (2)	3.90 (2)	3.90 (2)
-DSA Variable Fan Speed Control Package							
208–230/1/60, 277/1/60							
Fan RPM	1490	1490	1490	1490	N/A	N/A	N/A
Motor HP (Qty)	1.23 (1)	1.23 (2)	1.23 (2)	1.23 (2)	N/A	N/A	N/A
208–230/3/60							
Fan RPM	N/A	1490 (1) 1445 (1)	1490 (1) 1445 (1)	1490 (1) 1445 (1)	1500 (1) 1585 (1)	1500 (1) 1585 (1)	1500 (1) 1585 (1)
-DEC Intelligent Control Package with EC Fans							
208–230/1/60, 277/1/60							
Fan RPM	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Motor HP (Qty)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
208–230/3/60							
Fan RPM	1450	1450	1450	1450	1500	1500	1500
Motor HP (Qty)	1.39 (1)	1.39 (2)	1.39 (2)	1.39 (2)	3.85 (2)	3.85 (2)	3.85 (2)
460/3/60, 575/3/60							
Fan RPM	1440	1440	1440	1440	1500	1500	1500
Motor HP (Qty)	1.41 (1)	1.41 (2)	1.41 (2)	1.41 (2)	3.85 (2)	3.85 (2)	3.85 (2)
Motor HP (Qty)	N/A	1.23 (1) 1.58 (1)	1.23 (1) 1.58 (1)	1.23 (1) 1.58 (1)	3.85 (1) 3.58 (1)	3.85 (1) 3.58 (1)	3.85 (1) 3.58 (1)
460/3/60, 575/3/60							
Fan RPM	N/A	1440 (1) 1545 (1)	1440 (1) 1545 (1)	1440 (1) 1545 (1)	1500 (1) 1650 (1)	1500 (1) 1650 (1)	1500 (1) 1650 (1)
Motor HP (Qty)	N/A	1.41 (1) 1.76 (1)	1.41 (1) 1.76 (1)	1.41 (1) 1.76 (1)	3.85 (1) 3.90 (1)	3.85 (1) 3.90 (1)	3.85 (1) 3.90 (1)

SCS Air Cooled Condensers, Dual Circuit							
SCS	**312**	**447**	**525**	**597**	**683**	**940**	**1366**
-DAA Fan Cycling Control Package							
208–230/1/60, 277/1/60							
Fan RPM	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Motor HP (Qty)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
208–230/3/60							
Fan RPM	1585	1585	1585	1585	1585	1585	1585
Motor HP (Qty)	3.58 (2)	3.58 (2)	3.58 (3)	3.58 (3)	3.58 (3)	3.58 (4)	3.58 (6)
460/3/60, 575/3/60							
Fan RPM	1650	1650	1650	1650	1650	1650	1650
Motor HP (Qty)	3.90 (2)	3.90 (2)	3.90 (3)	3.90 (3)	3.90 (3)	3.90 (4)	3.90 (6)
-DSA Variable Fan Speed Control Package							
208–230/1/60, 277/1/60							
Fan RPM	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Motor HP (Qty)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
208–230/3/60							
Fan RPM	1500 (1) 1585 (1)	1500 (1) 1585 (1)	1500 (1) 1585 (2)	1500 (1) 1585 (2)	1500 (1) 1585 (2)	1500 (1) 1585 (3)	1500 (2) 1585 (4)
Motor HP (Qty)	3.85 (1) 3.58 (1)	3.85 (1) 3.58 (1)	3.85 (1) 3.58 (2)	3.85 (1) 3.58 (2)	3.85 (1) 3.58 (2)	3.85 (1) 3.58 (3)	3.85 (2) 3.58 (4)
460/3/60, 575/3/60							
Fan RPM	1500 (1) 1650 (1)	1500 (1) 1650 (1)	1500 (1) 1650 (2)	1500 (1) 1650 (2)	1500 (1) 1650 (2)	1500 (1) 1650 (3)	1500 (2) 1650 (4)
Motor HP (Qty)	3.85 (1) 3.90 (1)	3.85 (1) 3.90 (1)	3.85 (1) 3.90 (2)	3.85 (1) 3.90 (2)	3.85 (1) 3.90 (2)	3.85 (1) 3.90 (3)	3.85 (2) 3.90 (4)
-DEC Intelligent Control Package with EC Fans							
208–230/1/60, 277/1/60							
Fan RPM	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Motor HP (Qty)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
208–230/3/60							
Fan RPM	1500	1500	1500	1500	1500	1500	1500
Motor HP (Qty)	3.85 (2)	3.85 (2)	3.85 (3)	3.85 (3)	3.85 (3)	3.85 (4)	3.85 (6)
460/3/60, 575/3/60							
Fan RPM	1500	1500	1500	1500	1500	1500	1500
Motor HP (Qty)	3.85 (2)	3.85 (2)	3.85 (3)	3.85 (3)	3.85 (3)	3.85 (4)	3.85 (6)

10. Electrical Data

10.1. No Receiver Skid, Single and Dual Circuit

10.1.1. AA Models

SCS, AA Models, No Receiver Skid, Single and Dual Circuit										
Model	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-012-()AA	2.0	15	1.8	15	N/A		0.4	15	N/A	
SCS-018-()AA	2.0	15	1.8	15	N/A		0.4	15	N/A	
SCS-024-()AA	2.0	15	1.8	15	N/A		0.4	15	N/A	
SCS-036-()AA	5.4	15	4.9	15	3.8	15	2.0	15	1.6	15
SCS-060-()AA	5.4	15	4.9	15	3.8	15	2.0	15	1.6	15
SCS-096-()AA	10.7	15	9.7	15	7.7	15	4.1	15	3.3	15
SCS-120-()AA	10.7	15	9.7	15	7.7	15	4.1	15	3.3	15
SCS-144-()AA	10.7	15	9.7	15	7.7	15	4.1	15	3.3	15
SCS-192-()AA	N/A		N/A		22.7	25	12.3	15	9.9	15
SCS-252-()AA	N/A		N/A		22.7	25	12.3	15	9.9	15
SCS-276-()AA	N/A		N/A		22.7	25	12.3	15	9.9	15
SCS-312-()AA	N/A		N/A		22.7	25	12.3	15	9.9	15
SCS-447-()AA	N/A		N/A		22.7	25	12.3	15	9.9	15
SCS-525-()AA	N/A		N/A		34.1	35	18.5	20	14.8	15
SCS-597-()AA	N/A		N/A		34.1	35	18.5	20	14.8	15
SCS-683-()AA	N/A		N/A		34.1	35	18.5	20	14.8	15
SCS-940-()AA	N/A		N/A		45.4	50	24.7	25	19.7	20
SCS-1366-()AA	N/A		N/A		68.2	70	37.0	40	29.6	30

Key:
MCA = Minimum Circuit Amps
MFS = Maximum Fuse Size
FLA = Full Load Amps, listed per component on unit nameplates
Format = Voltage/Number of Phases/Frequency. E.g., 277/1/60 = 277 V / 1 phase / 60 Hz

10.1.2. SA Models

SCS, SA Models, No Receiver Skid, Single and Dual Circuit										
Model	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-012-( )SA	2.0	15	1.8	15	N/A		N/A		N/A	
SCS-018-( )SA	2.0	15	1.8	15	N/A		N/A		N/A	
SCS-024-( )SA	2.0	15	1.8	15	N/A		N/A		N/A	
SCS-036-( )SA	5.4	15	4.9	15	N/A		N/A		N/A	
SCS-060-( )SA	5.4	15	4.9	15	N/A		N/A		N/A	
SCS-096-( )SA	10.7	15	8.9	15	9.2	15	4.9	15	3.9	15
SCS-120-( )SA	10.7	15	8.9	15	9.2	15	4.9	15	3.9	15
SCS-144-( )SA	10.7	15	8.9	15	9.2	15	4.9	15	3.9	15
SCS-192-( )SA	N/A		N/A		23.6	30	12.3	15	9.8	15
SCS-252-( )SA	N/A		N/A		23.6	30	12.3	15	9.8	15
SCS-276-( )SA	N/A		N/A		23.6	30	12.3	15	9.8	15
SCS-312-( )SA	N/A		N/A		23.6	30	12.3	15	9.8	15
SCS-447-( )SA	N/A		N/A		23.6	30	12.3	15	9.8	15
SCS-525-( )SA	N/A		N/A		35.0	40	18.4	20	14.7	15
SCS-597-( )SA	N/A		N/A		35.0	40	18.4	20	14.7	15
SCS-683-( )SA	N/A		N/A		35.0	40	18.4	20	14.7	15
SCS-940-( )SA (1X4)	N/A		N/A		46.3	50	24.6	25	19.7	20
SCS-940-( )SA (2X2)	N/A		N/A		47.2	50	24.5	25	19.6	20
SCS-1366-( )SA	N/A		N/A		70.0	70	36.9	40	29.5	30

10.2. No Receiver Skid, Single and Dual Circuit

10.2.1. EC Models

SCS, EC Models, No Receiver Skid, Single and Dual Circuit										
Model	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-012-()EC	3.6	15	2.7	15	N/A		N/A		N/A	
SCS-018-()EC	3.6	15	2.7	15	N/A		N/A		N/A	
SCS-024-()EC	3.6	15	2.7	15	N/A		N/A		N/A	
SCS-036-()EC	N/A		N/A		4.0	15	2.9	15	2.3	15
SCS-060-()EC	N/A		N/A		4.0	15	2.9	15	2.3	15
SCS-096-()EC	N/A		N/A		8.0	15	5.8	15	4.6	15
SCS-120-()EC	N/A		N/A		8.0	15	5.8	15	4.6	15
SCS-144-()EC	N/A		N/A		8.0	15	5.8	15	4.6	15
SCS-192-()EC	N/A		N/A		24.5	30	12.2	15	9.7	15
SCS-252-()EC	N/A		N/A		24.5	30	12.2	15	9.7	15
SCS-276-()EC	N/A		N/A		24.5	30	12.2	15	9.7	15
SCS-312-()EC	N/A		N/A		24.5	30	12.2	15	9.7	15
SCS-447-()EC	N/A		N/A		24.5	30	12.2	15	9.7	15
SCS-525-()EC	N/A		N/A		36.8	40	18.3	20	14.6	15
SCS-597-()EC	N/A		N/A		36.8	40	18.3	20	14.6	15
SCS-683-()EC	N/A		N/A		36.8	40	18.3	20	14.6	15
SCS-940-()EC	N/A		N/A		49.0	50	24.3	25	19.5	20
SCS-1366-()EC	N/A		N/A		73.6	80	36.5	40	29.2	30

10.3. With Receiver Skid, Single Circuit

10.2.2. SAA Models

SCS, AA Models, With Receiver Skid, Single Circuit										
Model	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-012-SAA	2.7	15	2.4	15	N/A		0.8	15	N/A	
SCS-018-SAA	2.7	15	2.4	15	N/A		0.8	15	N/A	
SCS-024-SAA	2.7	15	2.4	15	N/A		0.8	15	N/A	
SCS-036-SAA	6.1	15	5.4	15	4.6	15	2.4	15	1.9	15
SCS-060-SAA	6.1	15	5.4	15	4.6	15	2.4	15	1.9	15
SCS-096-SAA	11.5	15	10.3	15	8.4	15	4.4	15	3.6	15
SCS-120-SAA	11.5	15	10.3	15	8.4	15	4.4	15	3.6	15
SCS-144-SAA	11.5	15	10.3	15	8.4	15	4.4	15	3.6	15
SCS-192-SAA	N/A		N/A		23.5	30	12.7	15	10.1	15

10.3.1. SSA Models

SCS, SA Models, With Receiver Skid, Single Circuit										
Model	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-012-SSA	2.7	15	2.4	15	N/A		N/A		N/A	
SCS-018-SSA	2.7	15	2.4	15	N/A		N/A		N/A	
SCS-024-SSA	2.7	15	2.4	15	N/A		N/A		N/A	
SCS-036-SSA	6.1	15	5.4	15	N/A		N/A		N/A	
SCS-060-SSA	6.1	15	5.4	15	N/A		N/A		N/A	
SCS-096-SSA	11.5	15	9.4	15	9.9	15	5.3	15	4.2	15
SCS-120-SSA	11.5	15	9.4	15	9.9	15	5.3	15	4.2	15
SCS-144-SSA	11.5	15	9.4	15	9.9	15	5.3	15	4.2	15
SCS-192-SSA	N/A		N/A		24.4	30	12.6	15	10.1	15

10.3.2. SEC Models

SCS, EC Models, With Receiver Skid, Single Circuit										
Model	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-012-SEC	4.3	15	3.3	15	N/A		N/A		N/A	
SCS-018-SEC	4.3	15	3.3	15	N/A		N/A		N/A	
SCS-024-SEC	4.3	15	3.3	15	N/A		N/A		N/A	
SCS-036-SEC	N/A		N/A		4.8	15	3.2	15	2.6	15
SCS-060-SEC	N/A		N/A		4.8	15	3.2	15	2.6	15
SCS-096-SEC	N/A		N/A		8.8	15	6.1	15	4.9	15
SCS-120-SEC	N/A		N/A		8.8	15	6.1	15	4.9	15
SCS-144-SEC	N/A		N/A		8.8	15	6.1	15	4.9	15
SCS-192-SEC	N/A		N/A		25.3	30	12.5	15	10.0	15

10.4. With Receiver Skid, Dual Circuit

10.4.1. DAA Models

SCS, AA Models, With Receiver Skid, Dual Circuit										
Model	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-060-DAA	6.9	15	6.0	15	5.3	15	5.3	15	5.3	15
SCS-096-DAA	12.2	15	10.8	15	9.2	15	9.2	15	9.2	15
SCS-120-DAA	12.2	15	10.8	15	9.2	15	9.2	15	9.2	15
SCS-144-DAA	12.2	15	10.8	15	9.2	15	9.2	15	9.2	15
SCS-192-DAA	N/A		N/A		24.2	30	13.0	15	10.4	15
SCS-252-DAA	N/A		N/A		24.2	30	13.0	15	10.4	15
SCS-276-DAA	N/A		N/A		24.2	30	13.0	15	10.4	15
SCS-312-DAA	N/A		N/A		24.2	30	13.0	15	10.4	15
SCS-447-DAA	N/A		N/A		24.2	30	13.0	15	10.4	15
SCS-525-DAA	N/A		N/A		35.6	40	19.2	20	15.4	20
SCS-597-DAA	N/A		N/A		35.6	40	19.2	20	15.4	20
SCS-683-DAA	N/A		N/A		35.6	40	19.2	20	15.4	20
SCS-940-DAA	N/A		N/A		46.2	50	25.0	30	20.0	25
SCS-1366-DAA	N/A		N/A		69.7	70	37.7	40	30.2	35

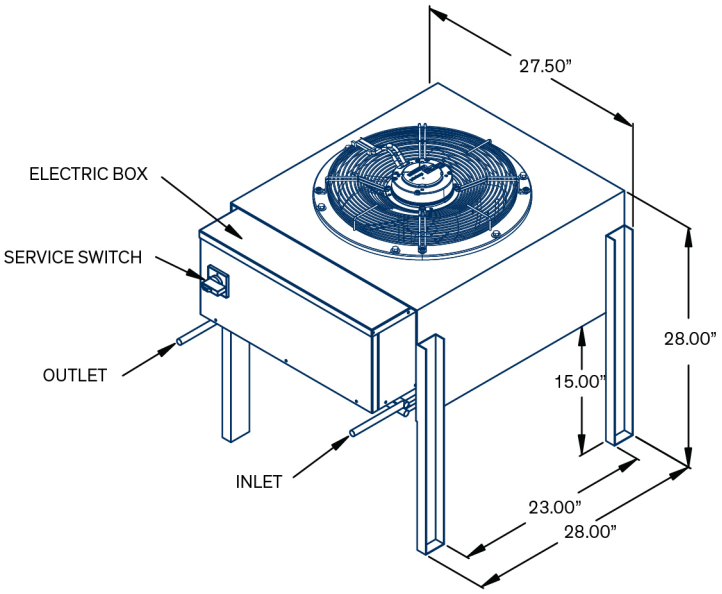
10.4.2. DSA Models

SCS, AA Models, With Receiver Skid, Dual Circuit										
Model	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-060-DSA	6.9	15	6.0	15	N/A		N/A		N/A	
SCS-096-DSA	12.2	15	10.0	15	10.7	15	10.7	15	10.7	15
SCS-120-DSA	12.2	15	10.0	15	10.7	15	10.7	15	10.7	15
SCS-144-DSA	12.2	15	10.0	15	10.7	15	10.7	15	10.7	15
SCS-192-DSA	N/A		N/A		25.1	30	12.9	15	10.4	15
SCS-252-DSA	N/A		N/A		25.1	30	12.9	15	10.4	15
SCS-276-DSA	N/A		N/A		25.1	30	12.9	15	10.4	15
SCS-312-DSA	N/A		N/A		25.1	30	12.9	15	10.4	15
SCS-447-DSA	N/A		N/A		25.1	30	12.9	15	10.4	15
SCS-525-DSA	N/A		N/A		36.5	40	19.1	20	15.3	20
SCS-597-DSA	N/A		N/A		36.5	40	19.1	20	15.3	20
SCS-683-DSA	N/A		N/A		36.5	40	19.1	20	15.3	20
SCS-940-DSA (1X4)	N/A		N/A		47.1	50	24.9	25	20.0	20
SCS-940-DSA (2X2)	N/A		N/A		48.0	50	24.9	25	19.9	20
SCS-1366-DSA	N/A		N/A		71.5	80	37.5	40	30.0	35

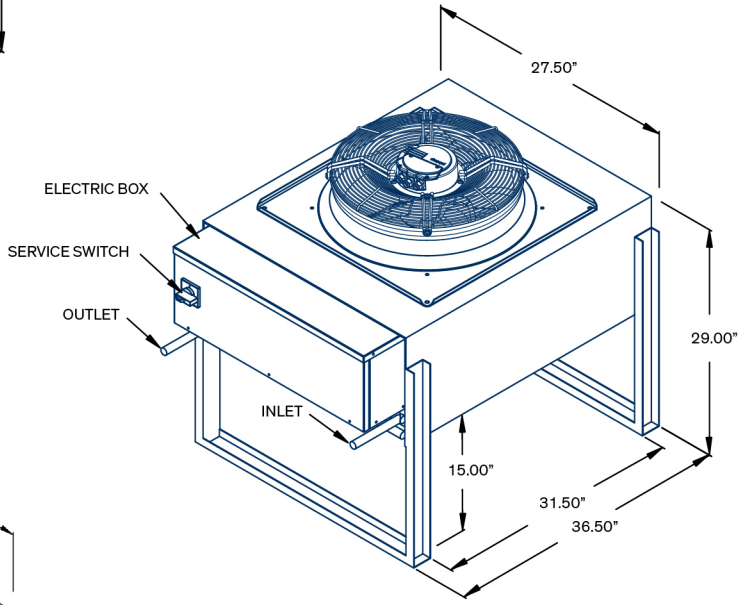
10.4.3. DEC Models

SCS, EC Models, With Receiver Skid, Dual Circuit										
Model	208–230/1/60		277/1/60		208–230/3/60		460/3/60		575/3/60	
	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS	MCA	MFS
SCS-060-DEC	N/A		N/A		5.5	15	5.5	15	5.5	15
SCS-096-DEC	N/A		N/A		9.5	15	9.5	15	9.5	15
SCS-120-DEC	N/A		N/A		9.5	15	6.4	15	5.2	15
SCS-144-DEC	N/A		N/A		9.5	15	6.4	15	5.2	15
SCS-192-DEC	N/A		N/A		26.0	30	12.8	15	10.3	15
SCS-252-DEC	N/A		N/A		26.0	30	12.8	15	10.3	15
SCS-276-DEC	N/A		N/A		26.0	30	12.8	15	10.3	15
SCS-312-DEC	N/A		N/A		26.0	30	12.8	15	10.3	15
SCS-447-DEC	N/A		N/A		26.0	30	12.8	15	10.3	15
SCS-525-DEC	N/A		N/A		38.3	40	18.9	20	15.2	20
SCS-597-DEC	N/A		N/A		38.3	40	18.9	20	15.2	20
SCS-683-DEC	N/A		N/A		38.3	40	18.9	20	15.2	20
SCS-940-DEC	N/A		N/A		49.8	50	24.7	25	19.8	20
SCS-1366-DEC	N/A		N/A		75.1	80	37.2	40	29.8	30

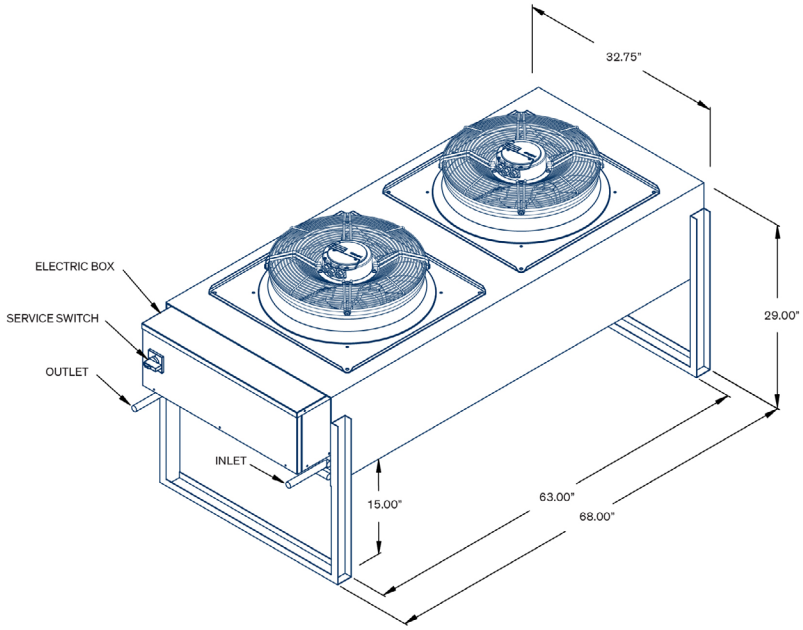
11. Dimensional Data



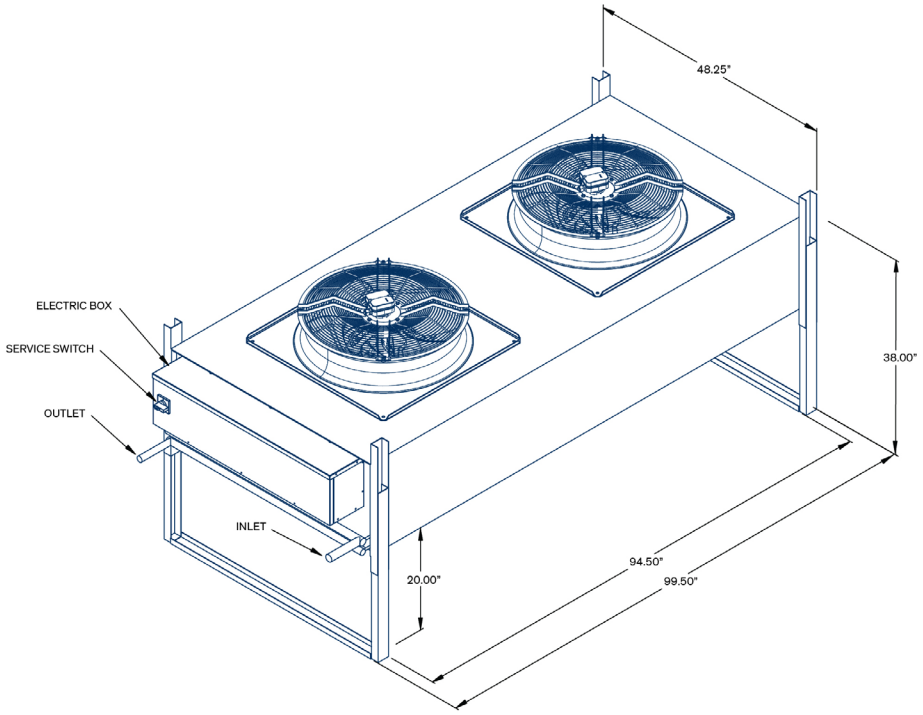
11.1. SCS-012 thru 024



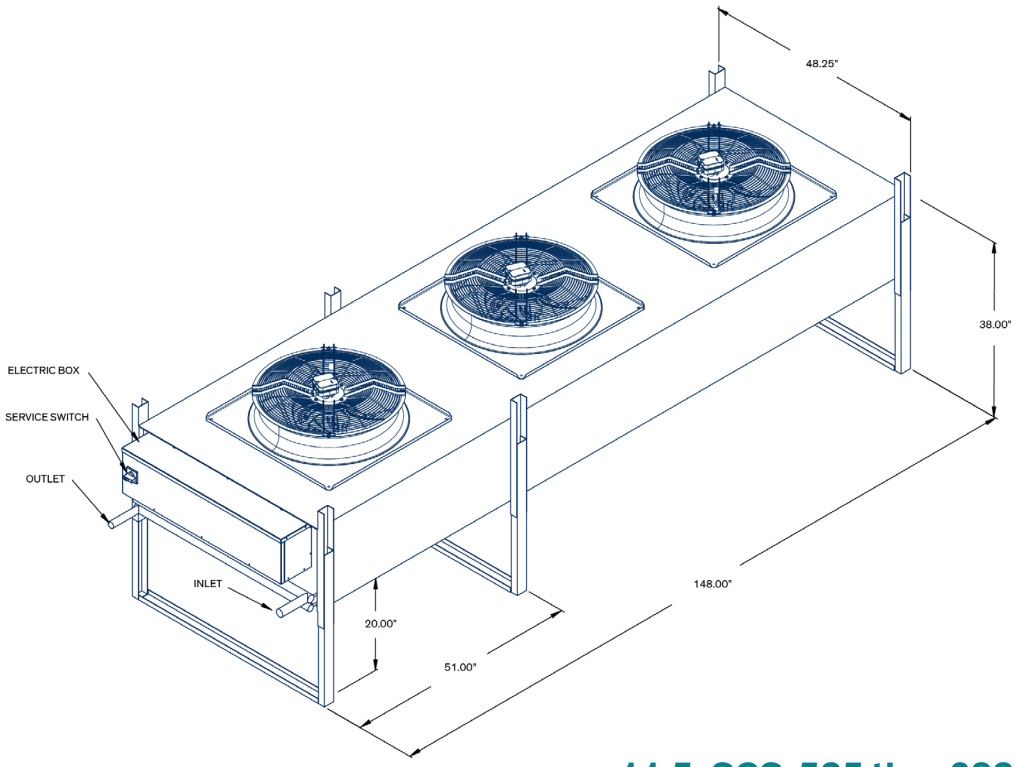
11.2. SCS-036 thru 060



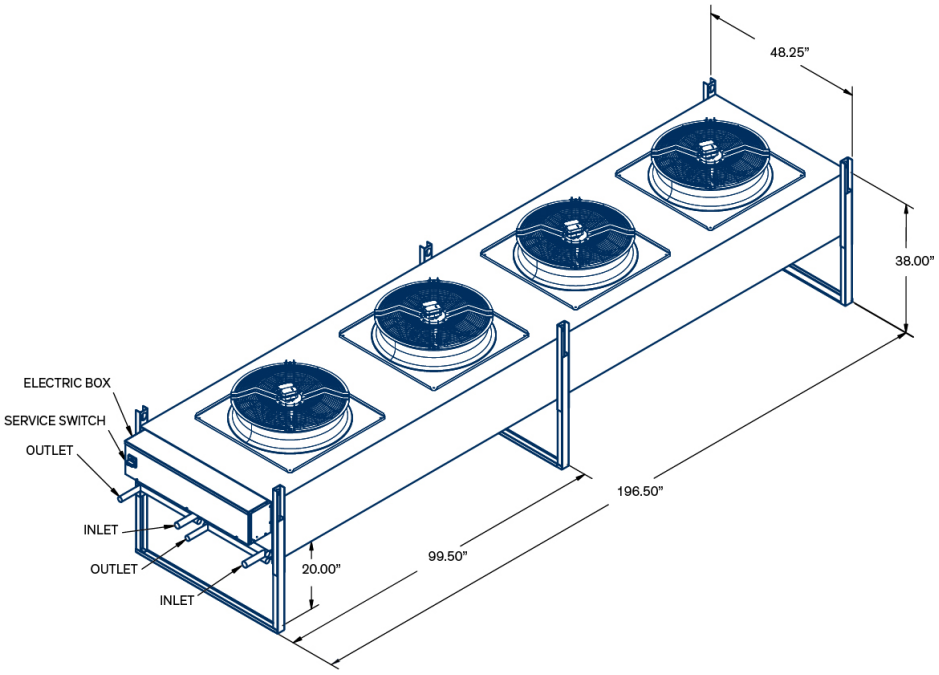
11.3. SCS-096 thru 144



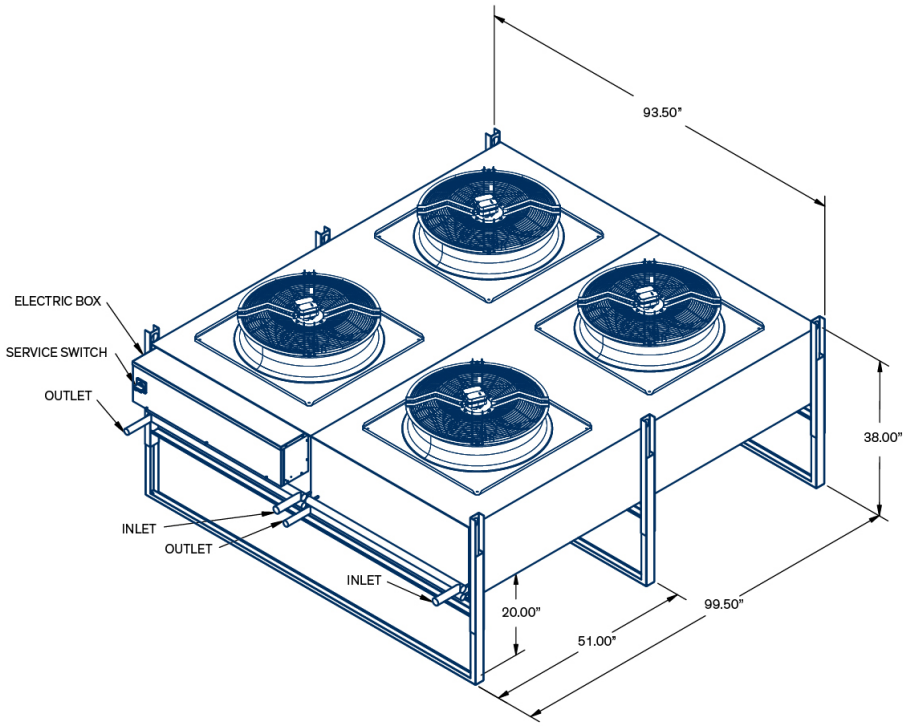
11.4. SCS-192 thru 447



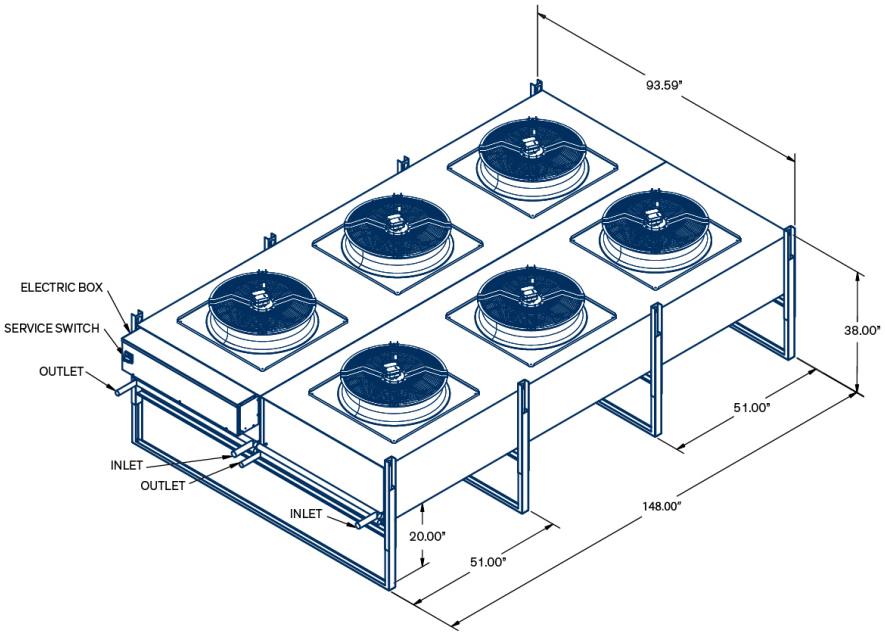
11.5. SCS-525 thru 683



11.6. SCS-940 1×4 Fan Configuration



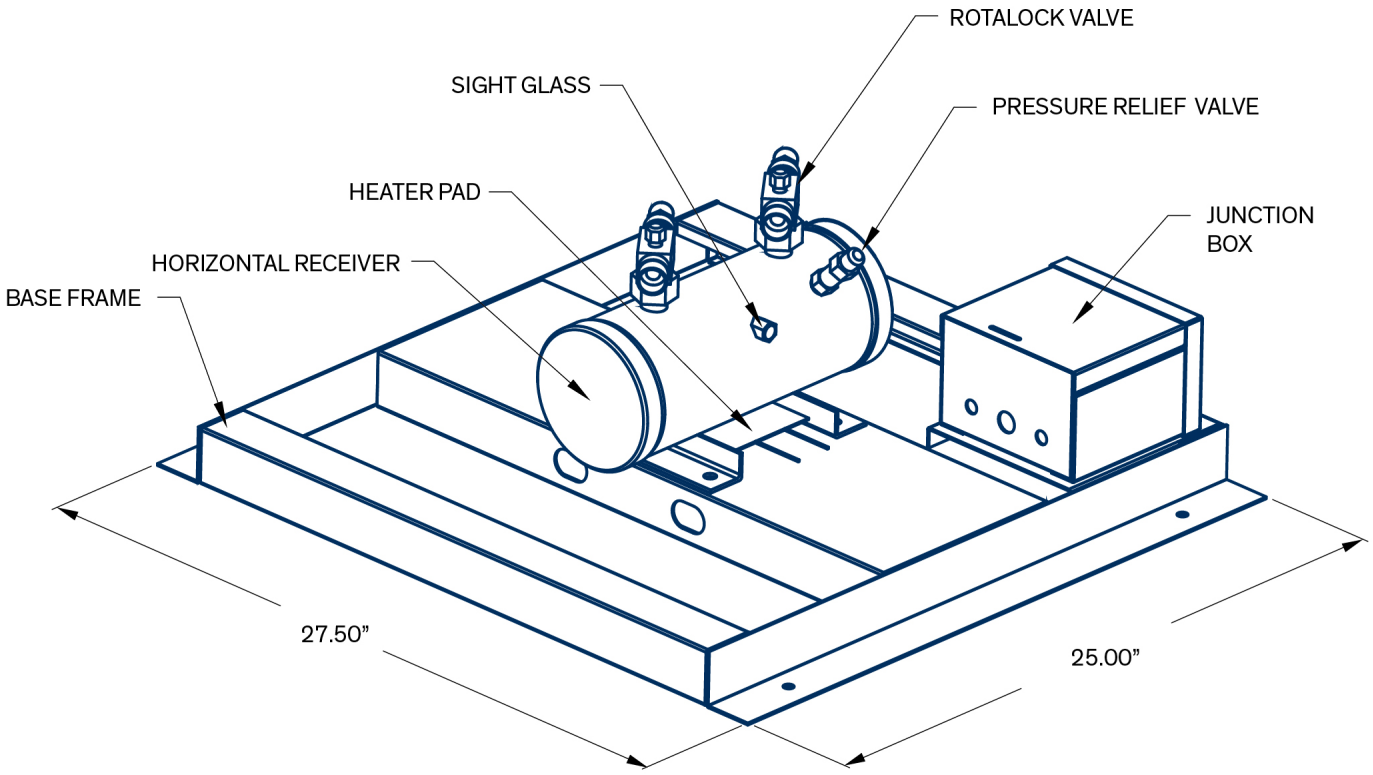
11.7. SCS-940 2×2 Fan Configuration


93.59"
ELECTRIC BOX
SERVICE SWITCH
OUTLET
INLET
OUTLET
INLET
38.00"
51.00"
148.00"
20.00"
51.00"

11.8. SCS-1366

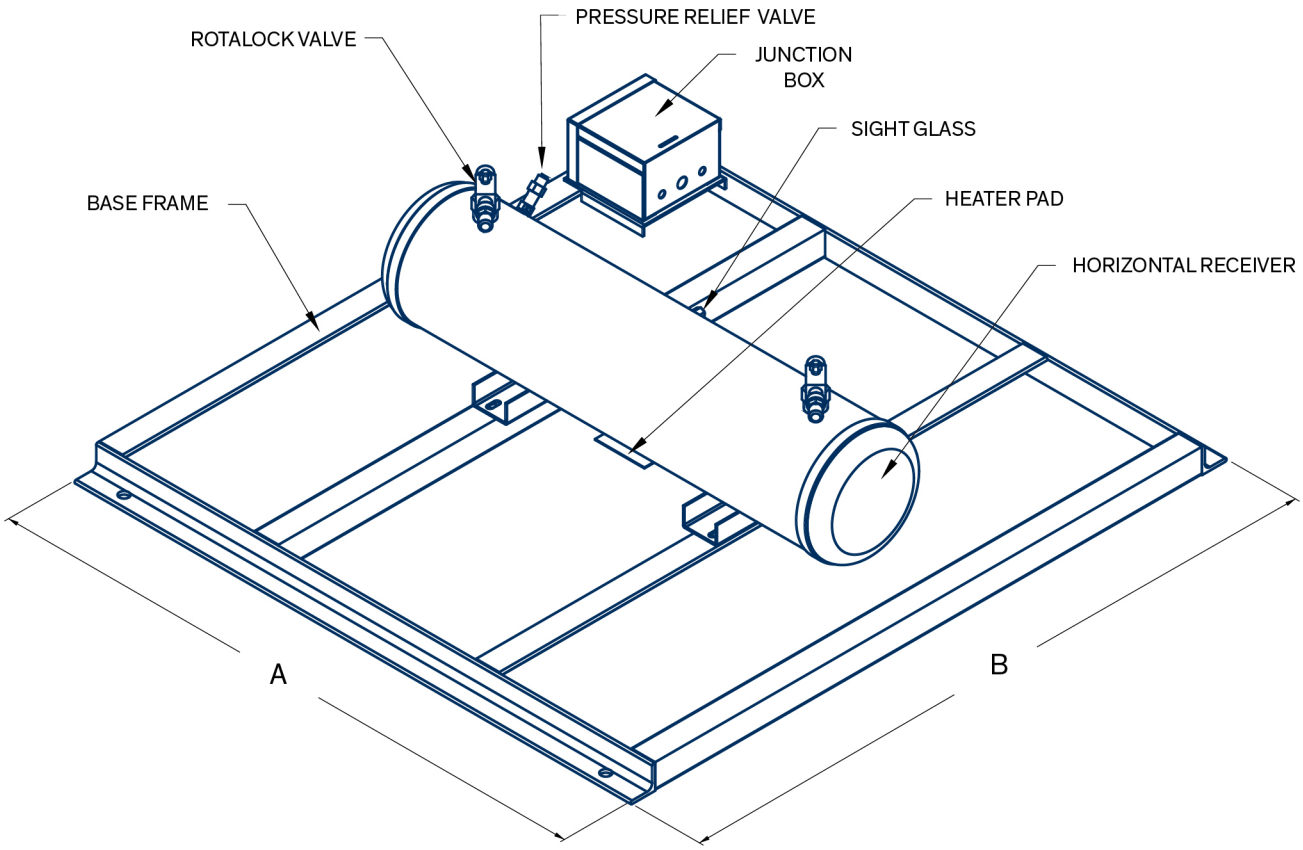
11.9. Receiver Skids, for -30°F Applications

Size	Used On	Weight (lb)
25" x 27.50"	SCS-012/024-S	45



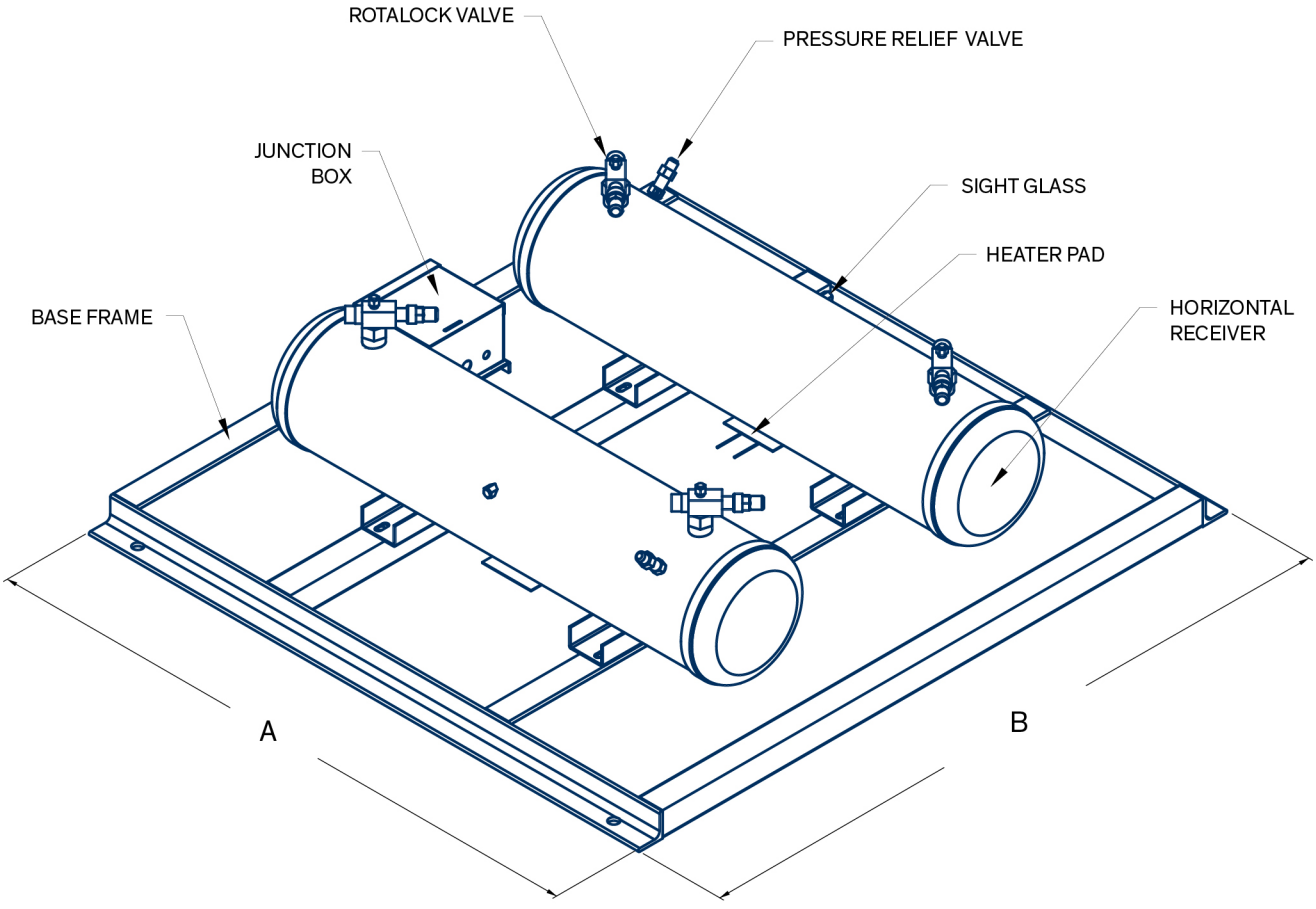
11.10. Receiver Skids, for -30°F Applications

A	B	Used On	Weight (lb)
30"	32.75"	SCS-036/096-S	55
30"	32.75"	SCS-120-S	60
30"	32.75"	SCS-144-S	85
45"	48.25"	SCS-252-S	90
45"	48.25"	SCS-276/312-S	110
45"	48.25"	SCS-447-S	115



11.11. Receiver Skids, for -30°F Applications

A	B	Used On	Weight (lb)
30"	32.75"	SCS-060/096-D	60
30"	32.75"	SCS-120/144-D	75
45"	48.25"	SCS-192/252-D	100
45"	48.25"	SCS-276/447-D	140
45"	48.25"	SCS-525/683-D	180
45"	48.25"	SCS-940-D	190
45"	48.25"	SCS-1366-D	240



12. Guide Specifications

SCS-XXX-XXX, Air-Cooled Condenser

12.1. Summary

This specification describes requirements for a refrigerant condensing system to be used with a DX based precision environmental control system. The STULZ SCS Condenser is an outdoor air-cooled condenser equipped with a direct-driven, external rotor-motor integrated in axial fans. The state-of-the-art SCS Condenser provides the total heat of rejection (THR) for the corresponding STULZ A/C evaporator units. The SCS Condenser is designed for both single and dual refrigeration circuits. The air-cooled condenser model number shall be SCS-XXX-XXX.

12.1.1. Design Requirements

The units are designed for outdoor installation with a removable front electric access panel. They do not require side access.

12.1.2. Refrigerant

Condensing units are designed for use with R407C or R410A refrigerant. The system is provided with a dry nitrogen charge and requires field evacuation and refrigerant charging.

12.1.3. Quality Assurance

STULZ maintains a set of international standards of quality management to ensure product quality. STULZ is ISO 9001 certified. Before shipment, each system is tested for complete operation and function.

12.2. Cabinet

The condenser cabinets are constructed of 0.080 in aluminum with heavy gauge galvanized steel support legs. The condenser cabinet houses the condenser coil, fans, fan guards, condenser motor control, and NEMA 3R electric box.

12.3. Head Pressure Control

Three options, must choose one: 0°F fan cycling, -20°F variable fan speed control, or -20°F intelligent control. And as an add on: -30°F flooded control with receiver.

All head pressure control options include refrigerant pressure transducers that monitor refrigerant discharge pressure then adjusts fans in response. Increased air flow over condenser coils enhances heat rejection, causing head pressure to decrease as cooling improves.

12.3.1. (AA) 0°F Fan Cycling Control (OHS only)

The unit cycles AC fans for low-ambient cooling control. This control maintains year-round air conditioning at a minimum ambient air temperature of 0°F DB. By cycling the condenser fans on refrigerant pressure, the refrigerant head pressure is maintained.

12.3.2. (SA) -20°F Variable Fan Speed Control

The unit blows a variable-speed fan for low-ambient cooling control, and cycles remaining fans as needed. This control maintains year-round air conditioning at a minimum ambient air temperature of -20°F DB. The speed of EC fan number one (closest to the header) is varied based on refrigerant discharge pressure. All remaining fans shall be controlled by fan cycling and are AC.

12.3.3. (EC) -20°F Intelligent Control

The unit blows variable-speed EC fans for low-ambient cooling control. This control maintains year-round air conditioning at a minimum ambient air temperature of -20°F DB. All fans are highly efficient, electronically commutated axial fans. All the EC fans are precisely controlled with variable speeds per pressure transducers to maintain refrigerant discharge pressure.

12.3.4. (_) -30°F Flooded Control with Receiver

The -30°F flooded control with receiver option is available in combination with any other head pressure control.

The unit runs a low-ambient flooded-head pressure control via a receiver and valve. This pressure control maintains year-round air conditioning at a minimum ambient air temperature of -30°F DB. This option includes an insulated liquid refrigerant receivers and parts described in sections 12.3.5 or 12.3.6, depending on the connected A/C unit model.

12.3.5. CeilAir Models (OHS)

The SCS condenser is provided for a ceiling mounted STULZ CeilAir A/C unit. Each CeilAir refrigerant circuit includes a factory installed crankcase heater, cold-start delay, liquid refrigerant receiver with receiver liquid-level sight glass, and head pressure control valve (for flooded condenser operation). The condenser includes branch circuit protection.

12.3.6. Floor Mounted Models (CFU/CFD, CRS, COS, CCU/CCD)

The SCS condenser is provided for a STULZ floor mounted A/C unit. The condenser is equipped with refrigerant pressure controls to cycle the condenser fans based on operating refrigerant pressure. Each refrigerant circuit includes a factory installed crankcase heater and cold-start delay for reliable low ambient operation. Each condenser fan is fixed-speed and includes internal overload protection.

For floor mounted models, the SCS condenser shall be provided with a receiver package which is factory preassembled on an aluminum mounting skid for easy installation below the SCS condenser or other serviceable location. Each receiver package includes an insulated liquid refrigerant receiver with liquid sight glass, Rotalock service valves (factory-installed on the receiver's inlet and outlet for refrigerant isolation), a pressure relief valve, and a receiver heater pad with internal thermostat control.

For floor mounted models, each refrigerant circuit includes a head pressure control valve. Field installation of the head pressure control valve is required. The condenser includes branch circuit protection and a power source for the receiver heater pads. Each receiver includes an electrical junction box for easy field wiring.

12.4. Mechanical Components

12.4.1. AC and EC Axial Fans

The fans shall be direct-driven, external rotor-motor integrated in an axial fan unit. The fan blades shall be constructed of a weather resistant, long-lasting coated steel or aluminum. Each fan is low noise and low vibration. Each fan impeller is dynamically and statically balanced in two planes to minimize vibration during operation.

12.4.2. Coil

The condenser coil is constructed of copper tubes with aluminum fins spaced no greater than 14 fins/in. Coils are factory pressure tested to UL 1995.

12.5. Electrical System

The electrical system conforms to National Electrical Code (NEC) class II requirements. The control circuit is 24 VAC. The control circuit wire is not smaller than 18 AWG. All wiring is neatly wrapped, run in conduit or cable trays, and routed in

bundles. Each wire ends with a service loop and is securely fastened per the NEC. Each wire is numbered for ease of service tracing.

All electrically actuated components are safely accessible from the unit's front without reaching over exposed high-voltage or rotating parts. Circuit breakers or manual motor starters protect each high-voltage circuit in all three phases. The blower motors have thermal and short circuit protection. Wiring is routed in separate bundles for line voltage and 24 V control circuits.

The electric box is conveniently located for full-service access. It includes all components required for unit operation, including contactors, starters, fuses, circuit breakers, and terminal boards.

12.5.1. Main Power Service Switch

The SCS condenser is mounted with a main power service switch.

12.6. Code Conformance

The supplied SCS Condenser system meets the following compliance approvals:

- c**ETL**us listed and conforms to UL standard 1995 (2015 Ed. 5)
- Certified to CSA standard C22.2 # 236 (2015 Ed. 5)
- Intertek control number 55789

13. Glossary

Term	Definition	Term	Definition
A	Amperes, SI unit of current	IAQ	Indoor Air Quality
AC	Alternating Current, in contrast to DC (direct current)	in. w.g.	Inches of Water Gauge, unit of pressure
AWG	American Wire Gauge, unit of wire diameter	IOM	Installation, Operation, and Maintenance manual
Btu/h	British Thermal Units per Hour	kW	Kilowatt, 1,000 watts
CCU/D	STULZ MiniSpace A/C system, upflow/downflow	LWT	Leaving Water Temperature
CFM	Cubic Feet per Minute, volume flow rate	MBH	1,000 Btu/h, rate of heat transfer
CFU/D	STULZ CyberAir A/C system, upflow/downflow	MCA	Minimum Circuit Ampacity, smallest wire size for main power
COS	STULZ CyberOne A/C system	MFS	Maximum Fuse Size, max amperage
CRAC	Computer Room Air Conditioner, often DX	NEC	National Electric Code, US standards
CRAH	Computer Room Air Handler, often CW	OHS	STULZ CeilAir Overheat System
CRS	STULZ CyberRow A/C system	PH	Phase, electrical current timing
CW	Chilled Water cooling method	psi	Pounds per Square Inch, unit of pressure
DB	Dry Bulb temperature, typical dry thermometer temperature without involving humidity, see WB	psig	Pounds per Square Inch Gauge Pressure, unit relative to atmospheric pressure
DF	Downflow airflow pattern, direction of airflow	SDS	Safety Data Sheet
EC	Electronically Commutated motor, variable speed	THL	Total Heat Load
DP	Dewpoint temperature	UF	Upflow airflow pattern, direction of airflow
DX	Direct Expansion cooling, refrigerant-based	UL	Underwriters Laboratories safety standards organization
°F	Degrees Fahrenheit, unit of temperature	UPS	Uninterruptible Power Supply
FLA	Full Load Amps, listed per component on unit nameplates	V	Volt, SI unit of electric potential
HP	Horsepower	VAC	Volts of Alternating Current
HVAC	Heating, Ventilation and Air Conditioning	VFD	Variable Frequency Drive
HX	Heat Exchanger, transfer heat between coolant circuits	W	Watts, SI unit of power
Hz	Hertz, unit of electrical frequency	WB	Wet Bulb temperature, wet thermometer temperature accounts for humidity, see DB

Voltage/Number of Phases/Frequency. 277/1/60 = 277V / 1 phase / 60 Hz

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